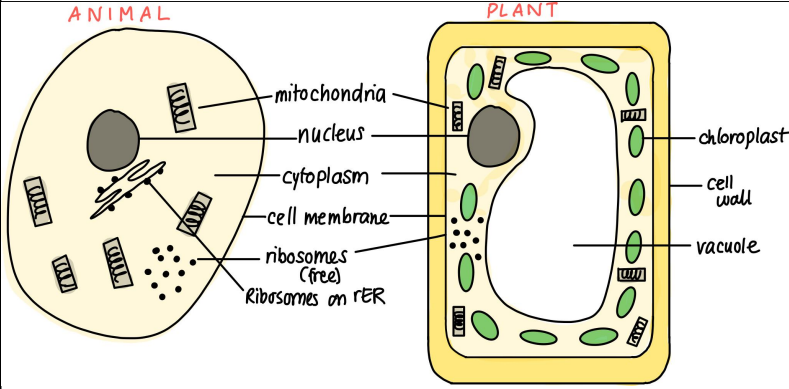
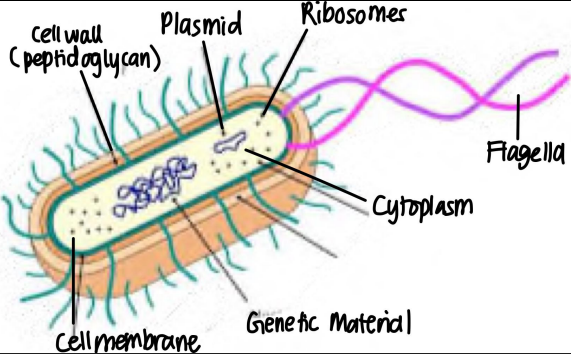


B1. CLASSIFICATION QUIZ CARDS

Question	Answer
List the characteristics of all living things	MRS GREN (movement, respiration, sensitivity, growth, reproduction, excretion, nutrition)
Define movement	an action by an organism or part of an organism causing a change of position or place
Define respiration	the chemical reactions in cells that break down nutrient molecules and release energy for metabolism
Define sensitivity	the ability to detect and sense stimuli in the internal or external environment and make appropriate responses
Define growth	A permanent increase in size and dry mass by an increase in cell number or cell size or both
Define reproduction	Processes that make more of the same kind of organism
Define excretion	Removal from organisms of the waste products of metabolism (chemical reactions in cells including respiration), toxic materials, and substances in excess of requirements
Define nutrition	taking in of materials for energy, growth and development; plants require light, carbon dioxide, water and ions; animals need organic compounds and ions and usually need water
Explain the difference between egestion and excretion	Excretion is the removal of metabolic waste. Egestion is food which has passed through the gut as faeces
Define and describe the binomial system	An internationally agreed system of naming species in which the scientific name of an organism is made up of two parts showing the genus and species
What is the scientific name for humans?	<i>Homo sapiens</i>
What is the traditional classification based on?	Developed by Carl Linnaeus and is based on morphology (shape/structure) and anatomy (internal structure)
What is classification now based on?	Sequences of bases in DNA and amino acids in proteins
What so classification today aim to show and how can this be presented?	Classification today aim to reflect on the evolutionary relationships that can be represented using a phylogenetic tree .
If organisms have a similar base sequence in DNA what does this tell you?	The organisms are more similar to each other than with those that they share a more distant ancestor
If organisms have a very different base sequence in DNA what does this tell you?	They are not closely related but may have a very distant common ancestor
List the classification system groups in order	Kingdom, Phylum, Class, Order, Family, Genus, Species
Define species	A group of organisms that can reproduce to produce fertile offspring
List the features in all cells	cytoplasm , cell membrane , DNA as genetic material, ribosomes for protein synthesis, enzymes involved in respiration
List the 5 kingdoms	Animal, plant, fungi, prokaryotes, protocists
Give the features and example of organisms in the animal kingdom	Cells do not have a cell wall. Multicellular and feed on other organisms. Example: tiger
Give the features and example of organisms in the plant kingdom	Multicellular organisms with cellulose cell wall. They have chloroplasts and carry out photosynthesis to produce glucose Example: ferns
Give the features and example of organisms in the fungi kingdom	Cells have chitin cell walls and reproduce using spores. Most fungi are saprotrophs, meaning they feed on dead decaying matter Example: yeast
Explain how a fungus is adapted to obtain its food [3]	It has hyphae that can branch out and grow on new food sources Has a large surface area It releases enzymes to break down the food
Give the features and example of organisms in the prokaryote kingdom	Have a cell wall made by peptidoglycan. Cells have no nucleus. Example: bacteria
Give the features and example of organisms in the protocists kingdom	Exist as single cells or colonies of single cells. Example: amoeba
Define invertebrates	Animals without a backbone

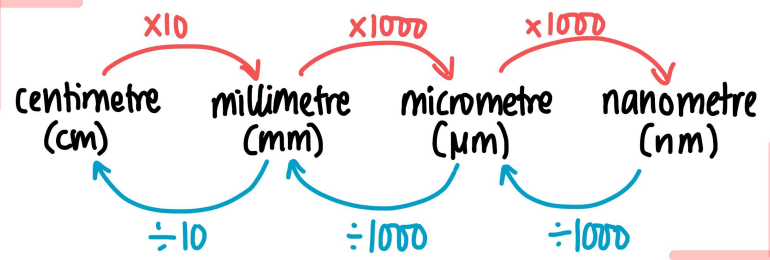
State the four groups of invertebrates and give an example for each	Nematodes (thin, tiny worms) Arthropods (insects, arachnids, crustaceans, myriapods) Annelids (earthworm) Molluscs (snail, clams)		
State two structural features of nematodes [2]	1. Unsegmented body 2. Thread-like bodies		
Define arthropods	A phylum that is part of invertebrates that have an exoskeleton, a segmented body, and jointed appendages (jointed legs)		
State the four groups of arthropods, listing the number of legs for each	Insects - 6 legs Arachnids - 8 legs Crustaceans - 10-14 legs Myriapods - >20 legs		
Define vertebrates	Animals with a backbone		
State the five groups of vertebrates, describing the features of each and giving an example	Vertebrate	Feature	Example
	Amphibians	moist permeable skin	frog
	Fish	wet scales, gills	goldfish
	Reptiles	dry scales	crocodile
	Birds	Feathers, beaks	sparrow
	Mammals	Fur/hair, mammary & sweat glands	rabbit
Describe the characteristics of viruses	Have protein coat and genes No mitochondria and no ribosomes so not considered living		
Describe the characteristics of ferns	Reproduce via spores Have no flowers or seeds They have stems and 'feathery' leaves		
Define flowering plants	Plants that have flowers and produce seeds Monocotyledons and Dicotyledons		
Compare and contrast monocotyledons and dicotyledons, giving examples	Monocotyledons		Dicotyledons
	Single cotyledon		Two cotyledons
	Long narrow leaf		Broad leaf
	Parallel veins		Networked veins
	Vascular Bundles Scattered		vascular bundles in a ring
	Flower parts x3		Flower parts x4 or x5
	E.g. grass		E.g. roses
	Define dichotomous keys	A series of paired statements/questions that lead to the identification of an organism	
Name some useful features of organisms that you could use in a dichotomous key	Useful features are ones that are easily identifiable from looking at the organism e.g. Numbers of legs, number of body segments, number of leaf segments, presence/absence of antennae etc... DO NOT CHOOSE - size (this will depend on the age of organism)		

B2. ORGANISATION OF THE ORGANISM QUIZ CARDS

Question	Answer
List the organelles of an animal cell	cell membrane, nucleus, cytoplasm, mitochondria, ribosomes, rough endoplasmic reticulum
List the organelles of a plant cell	cell membrane, nucleus, cytoplasm, mitochondria, ribosomes, rough endoplasmic reticulum, cellulose cell wall, vacuole, chloroplasts
Draw and label an animal and plant cell	
List the organelles of a prokaryote cell	Cell wall, cell membrane, DNA/genes, plasmids (loops of DNA), cytoplasm, ribosomes, some have a capsule, some have flagella
Draw and label a prokaryote cell	
Which structures are present in plant cells only?	Chloroplasts, cellulose cell wall, vacuole
Which structures are present in both plant and animal cells?	cell membrane, nucleus, cytoplasm, mitochondria, ribosomes, rough endoplasmic reticulum
Give the function of the cytoplasm	Where chemical reactions take place
Give the function of the nucleus	Contains genetic material and controls the activities of the cell
Give the function of the cell membrane	Controls what moves in and out of the cell e.g. oxygen, water, carbon dioxide.
Give the function of the cell wall	Provides support to the cell and strengthens the cell. Prevents the plant cell from bursting
What are plant cell walls made of?	Cellulose
What are cells walls of fungi made of?	Chitin
What are cell walls of bacteria made of?	Peptidoglycan
What is the function of the mitochondria?	Where aerobic respiration occurs to release energy
What is the function of the chloroplast?	Where photosynthesis takes place and where starch is stored
What is the function of ribosomes?	Protein synthesis
What is the function of the vacuole?	Stores cell sap (water salts & sugars) which maintains the shape of the cells
What is the function of the rough endoplasmic reticulum (rER)?	RER is a complex system of membranes covered in ribosomes to package and transport proteins made by the ribosomes
Why do cells with high rates of metabolism need a lot of mitochondria?	More mitochondria means more aerobic respiration occurs to release more energy
What are vesicles, where are they found and what are its functions?	Vesicles are small vacuoles surrounded by a membrane. They are found in the cytoplasm and they store and transport products.

Name 2 organelles that a bacteria, fungi, plant/algae, and animal cells all have	cytoplasm, cell membrane
Define cell	Building block of life
Define tissue	Group of cells with similar structures, working together to perform a shared function
Define organ	Group of tissues working together to perform specific functions
Define organ system	Group of organs with related functions, working together to perform body functions
Give an example of a cell, tissue and organ in a plant	palisade cell, palisade mesophyll tissue, leaf (organ)
Give an example of a cell, tissue, organ and organ system in an animal	e.g. muscle cell, muscular tissue, heart, circulatory system e.g. ciliated cell, epithelium tissue, lungs, respiratory system
Describe the functions and adaptations of ciliated cells	Function: Beat back and forth to move the mucus and trapped dust away from the lungs in the trachea Adaptation: cilia beat back and forth, contains a lot of mitochondria to release energy
Describe the functions and adaptations of root hair cells	Function: absorbs water (by osmosis) and mineral ions (by active transport) from the soil Adaptation: large surface area (due to long extensions) to absorb water and mineral ions
Describe the functions and adaptations of xylem vessels	Function: Transport water in plants and provides support to the leaf Adaptations: Dead and hollow cells with NO end walls - creates a continuous column. Lignin strengthens the cell
Describe the functions and adaptations of palisade mesophyll cells	Function: Absorb sunlight for photosynthesis to produce glucose Adaptations: Located in the upper surface of the leaf to absorb more sunlight & has many chloroplasts that contains chlorophyll to absorb light energy for photosynthesis
Describe the functions and adaptations of neurons	Function: Transmit electrical signals in the form of impulses Adaptations: Has long thin axons and dendrites. Myelinated sheath insulates the axon so that impulses can travel quicker
Describe the functions and adaptations of red blood cells	Function: Transports oxygen Adaptations: Red blood cells do NOT have a nucleus so there is more space for haemoglobin . It has a biconcave shape so there is a larger surface area for rapid exchange of oxygen. It has a flexible cell membrane so red blood cells can fit through narrow blood vessels (capillaries).
Describe the functions and adaptations of sperm cells	Function: fertilise the egg Adaptations: Has a flagellum to swim to the egg Has many mitochondria so more aerobic respiration can occur to release more energy Contains half of the genetic material (haploid nucleus) Acrosome contains digestive enzymes to digest the jelly coat of the egg
Describe the functions and adaptations of ovum (egg cells)	Function: To be fertilised by the sperm Adaptations: Large so it is easily found Contains half of the genetic material (haploid nucleus) Has a large cytoplasm that has an energy store Has a jelly coat that changes after fertilisation
Describe the functions and adaptations of lymphocytes	Function: Detect pathogens and produce antibodies Adaptations: Irregular shaped and can increase rapidly in numbers
Give the equation to calculate magnification	Magnification = image size/actual size
Describe the steps to approach and answer magnification questions	1. Write equation 2. Convert units 3. Substitute in values (with units) 4. State answer (with correct units)

Convert between cm, mm, micrometers and nanometres



B3. MOVEMENT IN AND OUT OF CELLS QUIZ CARDS

Question	Answer
Name three ways substances can move in and out of cells	Diffusion, osmosis, active transport
Define diffusion	The net movement of molecules or ions from an area of high concentration to an area of lower concentration, down a concentration gradient, across a partially permeable membrane until equilibrium is reached
What is a partially permeable membrane?	A membrane with small holes that allows some substances (like small water molecules) to pass through but will not allow larger molecules through.
What does concentration gradient mean?	The difference between the concentration of two locations e.g. inside a cell and outside a cell
What does net movement mean?	The overall / most common direction of movement
Give examples of diffusion	Oxygen diffuses into the bloodstream, CO ₂ diffuses into alveoli
Describe an example of diffusion in the lungs	There is a high concentration of oxygen in the alveoli of the lungs, which diffuses into the red blood cells in the bloodstream and can be transported around the body to respiring cells OR There is a high concentration of carbon dioxide in the bloodstream, which diffuses into the alveoli and can be exhaled as a waste product.
Describe an example of diffusion in plants	Oxygen and carbon dioxide diffuse down their concentration gradient between the atmosphere and the mesophyll of a plant through the stomata
Why do substances passively move between cells?	Because there is a concentration gradient
Where is the energy for diffusion obtained by?	Kinetic energy of RANDOM movement
Name 4 factors that affect the speed of diffusion	Temperature, size of molecules, distance over which to diffuse, steepness of the concentration gradient
Name 4 factors that speed up the rate of diffusion	A. Steeper concentration gradient B. Shorter distance C. Larger surface area D. Higher temperature
Define osmosis	The passive net movement of water from an area of high water potential (=dilute solution) to an area of lower water potential (=more concentrated solution), down a water potential gradient, across a partially permeable membrane until equilibrium is reached
Describe the water potential of a hypertonic solution compared to water potential of a cell	Hypertonic solution has relatively lower water potential (more concentrated)
Describe the water potential of an isotonic solution compared to water potential of a cell	Isotonic solution has the same water potential
Describe the water potential of a hypotonic solution compared to water potential of a cell	Hypotonic solution has relatively higher water potential (less concentrated)
Describe and explain what happens to a plant cell when placed in a hypertonic solution	Water moves out of the cell by osmosis, down a water potential gradient. The vacuole shrinks and cell membrane pulls away from the cell wall. The plant cell becomes flaccid;/plasmolysed
Describe and explain what happens to a plant cell when placed in an isotonic solution	Water molecules move but there is no net movement of water molecules
Describe and explain what happens to a plant cell when placed in a hypotonic solution	Water moves into the cell by osmosis, down a water potential gradient. The plant cell is turgid and cellulose cell wall prevents the cell from bursting
Describe and explain what happens to an animal cell when placed in a hypertonic solution	Water moves out of the cell by osmosis, down a water potential gradient. The animal cell shrinks
Describe and explain what happens to an animal cell when placed in an isotonic solution	Water molecules move but there is no net movement of water molecules
Describe and explain what happens to an animal cell when placed in a hypotonic solution	Water moves into the cell by osmosis, down a water potential gradient. The animal cell bursts (lysis) as there is no cell wall
Why are plant cells turgid?	Plants are supported by a turgor pressure within cells (water pressure acting against an inelastic cell wall)

Why is osmosis important in plants and animals?	In plants: water enter root hair cells by osmosis that leads to transpiration In animal: body cells can be damaged if they lose or gain too much water so osmosis controls this.
Plan an investigation to Investigate osmolarity in plant tissues	IV: Concentration of sucrose solution (0.0, 0.2, 0.4, 0.6, 0.8, 1.0 moldm ⁻³) --> This is prepared by mixing 1.0moldm ⁻³ sucrose solution with distilled water DV: The change in mass of the potato tuber tissue, units in grams. Percentage change in mass will be calculated using the formula: (final mass - initial mass)/initial mass x 100%. The mass will be measured using a balance CV: Same dimensions of potato cylinders (2cm in length - measured by rule) (use cork borer to keep the radius the same at 0.5cm) Same temperature of sucrose solution (keep in water bath set to 30°C and monitor with thermometer) Same duration the potato cylinders are kept in the sucrose solution (1 hour - use stopwatch) Same species of potato (use potatoes from 1 species only - maybe collected from same store/bag) Safety: cut the potato cylinders with a scapel carefully and cut away from fingers
Define active transport	The movement of molecules or ions from an area of low concentration to an area of higher concentration, against a concentratiion gradient, across a partially permeable membrane, using energy from respiration and carrier proteins
State the two requirements for active transport	Carrier proteins and energy from respiration
Give examples of active transport	Mineral ions across plant roots. Absorption of glucose by the epithelial cells of the villi in the small intestine
If too much water is in the soil around a plant, why can't it uptake mineral ions/nutrients?	Soil waterlogged, which means that there is no oxygen, so aerobic respiration cannot occur to produce energy for active transport
How do protein molecules move ions across a membrane during active transport	Proteins use energy from respiration to change the shape of the protein so ions move through the protein in the membrane, against a concentration gradient

B4. BIOLOGICAL MOLECULES QUIZ CARDS

Question	Answer
Give examples of biological molecules	Carbohydrates, Protein, fats, DNA
Give examples of a monosaccharide	Glucose
Give examples of a disaccharide	Sucrose
Give examples of a polysaccharide	Starch, cellulose, glycogen
Name 3 polysaccharides, where each is found and their function	Starch - plants - store of glucose Cellulose - plants - make cell walls Glycogen - animals - store of glucose
How is the structure of glycogen related to its function	Coiled & compact - good for storage Insoluble - Doesn't affect osmosis
How is the structure of starch related to its function	Coiled & compact - good for storage Insoluble - Doesn't affect osmosis
How is the structure of cellulose related to its function	Long, unbranched chains held together with many bonds = strong and rigid cell wall
What are proteins made of?	Amino acids
What are fats made of?	Fatty acids and glycerol
State the elements found in starch	Carbon (C), Hydrogen (H), Oxygen (O)
State the elements found in glucose	Carbon (C), Hydrogen (H), Oxygen (O)
State the elements found in fats	Carbon (C), Hydrogen (H), Oxygen (O)
State the elements found in proteins	Carbon (C), Hydrogen (H), Oxygen (O), Nitrogen (N), sometimes sulphur (S)
Describe the chemical test for starch and state the negative and positive result	Iodine solution Positive: turn blue black Negative: stay brown
Describe the chemical test for glucose (reducing sugars) and state the negative and positive result	Benedict's solution AND Heat to 70°C Positive: turn green, yellow, orange, red, brick red (increasing levels of sugar) Negative: stay blue
Describe the chemical test for fats and state the negative and positive result	Emulsion test: add ethanol and water and shake Positive: turn cloudy/milky white Negative: stay same
Describe the chemical test for vitamin C and state the negative and positive result	DCPIP solution Positive: turn clear Negative: stay same (blue)
Describe the chemical test for proteins and state the negative and positive result	Biuret solution Positive: turn purple Negative: stay same (blue)
What determines the function of a protein molecule?	The different sequence of amino acids results in different shapes to the protein molecules. The shape/structure determine the function
Give examples of what is made of protein	Enzymes, antibodies, tendons (join bones to bones), haemoglobin
What does DNA stand for?	Deoxyribose Nucleic Acid
What is a nucleotide made up of?	deoxyribose pentose sugar, phosphate, base
Describe the structure of DNA	A double helix made up of 2 nucleotides coiled together. The cross links between strands are formed by pairs of bases.
What are the 4 bases in DNA?	A, T, C, G
Describe complementary base pairing	A binds with T with 2 hydrogen bonds, G binds with C with 3 hydrogen bonds
What is the monomer of DNA?	Nucleotides
What is the chemical formula for water?	H ₂ O
Explain the importance of water	Acts as a solvent to allow * Digestion - food broken down by digestive enzymes dissolved in water * Excretion - excess waste dissolved in urine * Transport - dissolved substances (solutes) transported in blood plasma

B5. ENZYMES QUIZ CARDS

Question	Answer
Define catalyst	A substance that increases the rate of a chemical reaction and is NOT changed by the reaction
Define enzyme	A protein that is a biological catalyst that speeds up the rate of reaction without being used up itself
Why are enzymes important?	Speed up reactions in living organisms to sustain life
Explain why enzymes are necessary for biological processes [3]	They act as a catalyst to speed up reactions to sustain life as without enzymes, reactions happen too slowly. Enzymes can also reduce the activation energy .
Name 3 types of reactions enzymes can catalyse	Break large molecules into small ones, build up large molecules from smaller ones, convert one small molecule into another small molecule
Describe the process of enzymes catalysing a reaction	An enzyme has an active site, which has a specific shape and charge that is complementary to the substrate. A successful collision between the active site and substrate forms an enzyme-substrate complex and the products are released from active site. The enzyme is unchanged and can be used again
Why can enzymes only catalyse one reaction?	The active site of an enzyme has a specific shape and charge that is complementary to one substrate
Describe the lock and key model	The enzyme's active site has a specific shape (like a lock) that is complementary to the shape of the substrate (like a key)
What are enzymes made of?	Protein, which is made up of amino acids
Name 2 factors that affect enzyme activity	Temperature and pH
Describe how temperature affects enzyme activity	* At low temperatures, enzyme and substrate move slowly so there is low chance of successful collision * As temperature increases, kinetic energy increases and the substrate and enzyme move faster so this increases the chance of successful collision * At the optimum temperature = maximum enzyme activity * As temperature further increases, hydrogen bonds holding the enzyme together breaks and the enzyme (and active site) changes shape --> denaturation
Describe how pH affects enzyme activity	* At the optimum pH, there is maximum enzyme activity * As pH deviate from the optimum, the charge on the enzyme changes and enzyme activity decreases * At extreme pH values --> active site changes shape --> denaturation
Define denaturation	Change in the shape of the active site of an enzyme, preventing the substrate from binding

B6. PLANT NUTRITION QUIZ CARDS

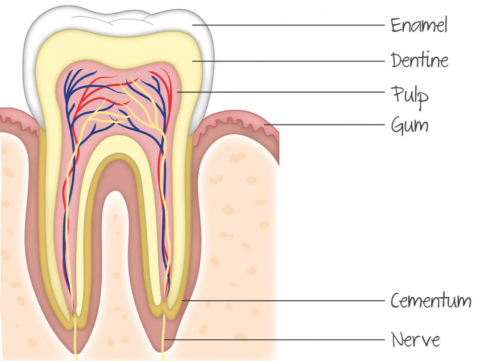
Question	Answer
Define photosynthesis	Water reacts with carbon dioxide in the presence of light to produce glucose and oxygen. It is the process by which plants manufacture carbohydrates from raw materials using energy from light
Where does photosynthesis occur?	In the chloroplasts in the cells of plants and algae
What is needed for photosynthesis?	light, chlorophyll, carbon dioxide, water
Give the word and balanced symbol equation for photosynthesis	Carbon dioxide + Water $\xrightarrow{\text{light and chlorophyll}}$ glucose + oxygen $6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{light and chlorophyll}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
Describe the role of chlorophyll in photosynthesis	Chlorophyll is a green pigment in chloroplasts that absorb light energy from sunlight and convert into chemical energy for the synthesis of carbohydrates
State the different uses of glucose	Respiration, starch (energy storage), cellulose cell wall, make proteins, fats and sugars
How does carbon dioxide get into a plant?	Through the stomata
What can glucose be stored as in a plant?	starch
Define limiting factors	factor present in the environment in such short supply that it restricts life processes
Give examples of the limiting factors of photosynthesis in different environments	Dull day in the summer: limiting factor = light Sunny day in the winter: limiting factor = temperature
Name 5 factors that can limit the rate of photosynthesis	Temperature, carbon dioxide concentration, light intensity, concentration of chlorophyll, volume of water
Explain how to destarch the plant	De-starch the plant = put in darkness for 48 hours before the experiment starts to ensure there is no starch in the plant
Describe an experiment to prove that light is needed for photosynthesis	De-starch plant $\rightarrow$ put a strip of aluminium foil on a part of the plant to block sunlight $\rightarrow$ put the plant under sunlight $\rightarrow$ test for starch using iodine solution Results: not covered by aluminium foil (= exposed to light) = starch present = photosynthesis occurred covered by aluminium foil (= NOT exposed to light) = starch absent = NO photosynthesis
Describe an experiment to prove that chlorophyll is needed for photosynthesis	Use a variegated leaf $\rightarrow$ De-starch plant $\rightarrow$ put the plant under sunlight $\rightarrow$ test for starch using iodine solution Results: Green parts (= have chlorophyll) = starch present = photosynthesis occurred White parts (NO chlorophyll) = starch absent = NO photosynthesis
Describe an experiment to prove that carbon dioxide is needed for photosynthesis	De-starch plant $\rightarrow$ put in bell jars (one containing sodium bicarbonate (i.e. baking soda) that produces CO_2 and the other containing sodium hydroxide (i.e. soda lime) that absorbs CO_2) $\rightarrow$ put the plant under sunlight $\rightarrow$ test for starch using iodine solution Results: Sodium bicarbonate (produce CO_2) = starch present = photosynthesis occurred Sodium hydroxide (absorb CO_2) = starch absent = NO photosynthesis
Describe the effect of increasing light intensity on the rate of photosynthesis	Low light intensities: LF=light, as light intensity increases, rate increases At high light intensities: LF = NOT light but other factors (e.g. water, temp, CO_2), as light intensity increases, rate constant (plateaus)
Describe the effect of increasing temperature on the rate of photosynthesis	Low temperature: LF=temperature, as temperature increases, rate increases At high temperature, enzymes denature $\rightarrow$ rate decreases rapidly
Describe the effect of increasing carbon dioxide concentration on the rate of photosynthesis	Low CO_2 conc.: LF= CO_2 , as CO_2 conc. increases, rate increases At high CO_2 conc.: LF = NOT CO_2 but other factors (e.g. water, temp, light), as CO_2 conc. increases, rate constant (plateaus)

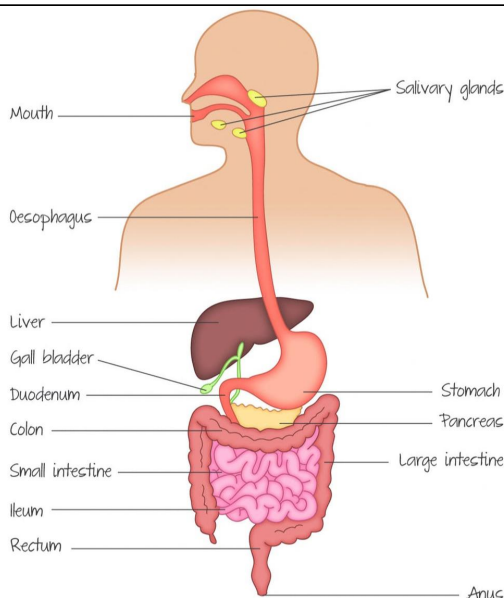
Describe the use of the hydrogen-carbonate indicator	Detects carbon dioxide produced (as it is an indicator that changes colour in different pH, and carbon dioxide is acidic when dissolved in solution) Highest conc of CO₂ = Yellow High = Orange Atmospheric level = Red Low = Magenta Lowest conc of CO₂ = Purple
A student was growing seedlings. Why did they put them in a clear plastic bag?	A. Transparent for light to get through & reduce water loss (transpiration)
How can you test if a plant contains starch?	A. Add iodine. It will turn from yellow/brown to blue/black
If an aquatic plant is producing bubbles, what do these bubbles contain?	Oxygen
How are leaves adapted for photosynthesis?	1. Large surface area - to absorb light 2. Thin - short diffusion pathway for gases can diffuses in and out 3. Many chloroplasts - absorb sunlight 4. Xylem and phloem - transport water/mineral ions/sugars to the parts of the plant that need them
Draw and label the parts of a cross-sectional diagram of a leaf	
Why are there more stomata on the lower side of the leaf than the upper?	To reduce the rate of transpiration (loss of water vapour by diffusion), as direct sunlight on the upper surface of the leaf raises temperatures, and therefore there is more water loss. Plants need to conserve water for photosynthesis
What controls the opening and closing of stomata?	Guard cells
Describe how guard cells cause the stomata to open	Ions diffuse into the guard cells, creating a water potential gradient. Water moves into the guard cells by osmosis, causing the guard cells to curve and the stomata to be open
Why do stomata often close at night?	No sunlight, so no photosynthesis, so carbon dioxide is not required to enter the plant from the atmosphere. Closing the stomata reduces water loss/transpiration
Why do stomata close in very hot, dry conditions?	To prevent a high rate of transpiration/water loss
When do plants carry out respiration? When do plants carry out photosynthesis?	Photosynthesis - in the daylight. Respiration - all the time! Plants are living!
A plant is able to photosynthesise, but the stomata are closed. Explain this	Carbon dioxide is also produced when the plant cells carry out respiration, this product of respiration can be used as a reactant of photosynthesis
Describe the function and adaptations of chloroplasts	Contains chlorophyll for photosynthesis
Describe the function and adaptations of waxy cuticle	Stops water evaporating
Describe the function and adaptations of guard cells	Open/closes the stomata to allow diffusion (CO ₂ in and O ₂ out). Contains chloroplasts
Describe the function and adaptations of stomata	Controls movement of gases in and out for photosynthesis and respiration. Controls the rate of diffusion Allows carbon dioxide to diffuse by diffusion into the leaf Allows oxygen to diffuse by diffusion out of the leaf
Describe the function and adaptations of upper/lower epidermis	Protect inner cells No chloroplasts

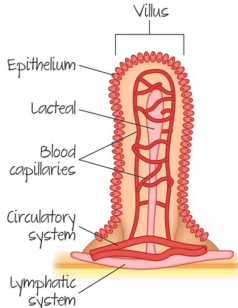
Describe the function and adaptations of palisade mesophyll	Closely packed - have many chloroplasts to allow photosynthesis
Describe the function and adaptations of spongy mesophyll	Loosely spaced to allow the diffusion of gases
Describe the function and adaptations of vascular bundle	Contains xylem and phloem vessels
Describe the function and adaptations of xylem	Transport water from roots to the leaf
Describe the function and adaptations of phloem	Transport sucrose from source to sink
State the 2 mineral ions important for plant growth	Magnesium and nitrate ions
State the use of magnesium ions	Make chlorophyll in chloroplasts for photosynthesis
State the use of nitrate ions	Make amino acids to make proteins
Explain the effect of magnesium ion deficiency on plant growth	Chlorosis (yellow leaves)
Explain how magnesium deficiency affects the growth and appearance of this plant [3]	Magnesium require to make chlorophyll so without magnesium, there is no chlorophyll so the leaves are yellow . This means less photosynthesis occurs and less glucose is produced so less growth .
Explain the effect of nitrate ion deficiency on plant growth	Plants absorb nitrate ions Nitrate ions are needed to make amino acids and proteins to make DNA Proteins are needed for growth so nitrate deficiency = stunted growth
Describe the features of glasshouses	CO2 enrichment Optimum light Optimum temperature

B7. HUMAN NUTRITION QUIZ CARDS

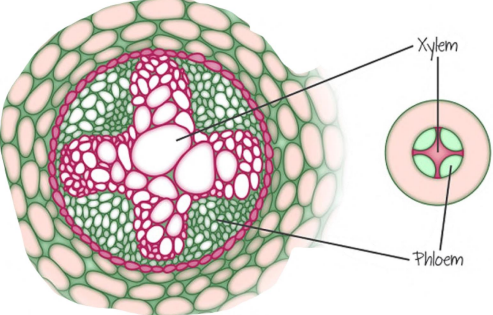
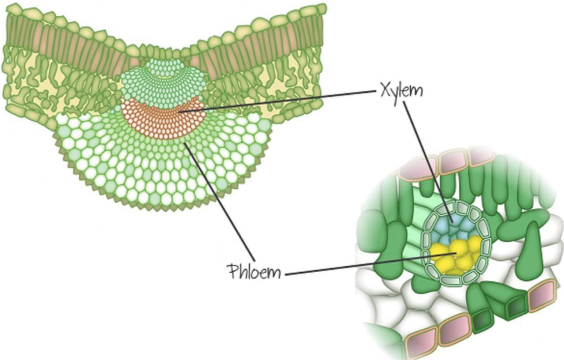
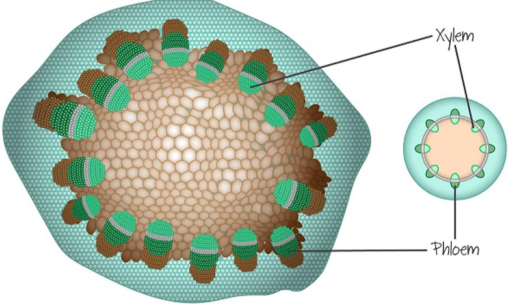
Question	Answer
Define balanced diet	Provides nutrients (carbohydrate, protein, fats, minerals, vitamins, fibre, water) in the correct quantities to provide sufficient energy . It also provides molecules for metabolism
Explain why the body requires carbohydrates , examples of food that contain it and the deficiency disease	Required to provide energy Examples: pasta, rice, bread, potato Deficiency disease: ketosis (increase blood acidity)
Explain why the body requires fats and oils , examples of food that contain it and the deficiency disease	Required to provide energy, thermal insulation, storage and protection of organs Examples: saturated fats (animal products like cheese), unsaturated fats (plant products like sunflower oil) Deficiency disease: increase in saturated fats will increase blood cholesterol and lead to coronary heart disease
What is the difference between saturated and unsaturated fats?	saturated - from animal products and can cause an increase in cholesterol in the blood unsaturated - from plant products e.g. oils and are less harmful to the body
Explain why the body requires proteins , examples of food that contain it and the deficiency disease	Required to provide energy (if in excess), growth and repair Examples: meat, fish, eggs, beans Deficiency disease: <u>Kwashiorkor</u> - swelling of the abdomen and legs, dry hair, flaky skin. <u>Marasmus</u> - very underweight, thin arms and legs, old-looking face
Describe the similarities and differences between marasmus and kwashiorkor [4]	Both protein deficiency Both deficiency disease Both give low body mass Both primarily affects children People with kwashiorkor have swelling of abdomen or legs People with marasmus are also deficient in carbohydrates/fats
Explain why the body requires vitamin C , examples of food that contain it and the deficiency disease	Required for collagen, resistance to diseases Examples: citrus fruits, lemon Deficiency disease: scurvy (bleeding gums)
Explain why the body requires vitamin D , examples of food that contain it and the deficiency disease	Required to strength bones and teeth Examples: eggs, fish oil Deficiency disease: rickets (legs bend outwards in children) osteomalacia (in adults - the softening of the bones and more likely to get fractures)
Explain why the body requires vitamin A , examples of food that contain it and the deficiency disease	Required for good vision Examples: eggs Deficiency disease: night blindness
Explain why the body requires calcium , examples of food that contain it and the deficiency disease	Required to strength bones and teeth Examples: milk, fish Deficiency disease: rickets, weak and brittle bones
Explain why the body requires iron , examples of food that contain it and the deficiency disease	Required to make haemoglobin to transport oxygen Examples: liver, meat, eggs Deficiency disease: anaemia (tired and lack of energy (as lack of haemoglobin to transport oxygen around the body)
Explain why the body requires fibre , examples of food that contain it and the deficiency disease	Required to add bulk to food Examples: grain, sweetcorn Deficiency disease: constipation
Explain why the body requires water , examples of food that contain it and the deficiency disease	Required as a solvent Examples: cucumber, water Deficiency disease: constipation, dehydration
Name 3 factors that affect dietary needs	Age, sex and activity level (lifestyle)
Why do women have a lower energy requirement than men?	Women have higher fat content, which has a lower metabolic rate than muscle
What is basal metabolic rate?	The energy required for chemical reactions in your body to be carried out when you are resting

Why would someone become obese?	Their energy input (in their food) is larger than energy output (in their movement/exercise), High intake of fatty food, high intake of sugary foods, too little exercise, social and emotional stress				
What happens due to starvation?	lack of fat, muscle wasting, lack of resistance to infections				
Define constipation	when the movement of food through the intestines slows down and it is hard to defecate				
Define ingestion	the taking of substances, e.g. food and drink, into the body through the mouth				
Define mechanical digestion	the breakdown of food into smaller pieces without chemical change to the food molecules				
Define chemical digestion	the breakdown of large, insoluble molecules into small, soluble molecules				
Define absorption	the movement of small food molecules and ions through the wall of the intestine into the blood				
Define assimilation	the movement of digested food molecules into the cells of the body where they are used, becoming part of the cells				
Define egestion	the passing out of food that has not been digested or absorbed, as faeces, through the anus				
Complete the table	Type of enzyme	Enzyme	Where is it produced ?	Substrates and Products	Where it works
	Carbohydrase	Amylase	Salivary glands Pancreas	Starch → Maltose	Mouth
		Maltase	ileum	Maltose → Glucose	ileum
		Sucrase	ileum	Sucrose → Glucose + Fructose	ileum
	Protease	Pepsin	stomach *	Proteins → amino acids	stomach
		Trypsin	Pancreas ileum		ileum
	Lipase	Lipase	Pancreas ileum	Lipids → fatty acids + glycerol	ileum
Draw and label a diagram of a tooth	 Enamel Dentine Pulp Gum Cementum Nerve				

Draw and label a diagram of the alimentary canal	 The diagram illustrates the human digestive system. Labels on the left side include: Mouth, Oesophagus, Liver, Gall bladder, Duodenum, Colon, Small intestine, Ileum, and Rectum. Labels on the right side include: Salivary glands, Stomach, Pancreas, Large intestine, and Anus.
State the function of each of the four types of teeth	Incisors - bite and cut Canine - pierce and tear Molars and Pre-molars - grind and chew
What are our first set of teeth called?	Milk teeth
Describe digestion in the mouth	Physical digestion by teeth and tongue Chemical digestion as salivary gland produces amylase to break down starch to glucose
Describe what happens in the oesophagus	Peristalsis - wave-like circular muscle contractions to move food from the mouth to the stomach
Describe what happens in the stomach	Physical digestion as walls contract churning food Chemical digestion as pepsin (a protease enzyme) breaks down protein into amino acids Hydrochloric acid kills bacteria and creates the optimum pH (pH2) for pepsin to work
What is the mixture of the contents in the stomach called?	Chyme (=HCL, pepsin and food mixed together)
Describe what happens in the liver	Produces bile
Describe what happens in the pancreas	Chemical digestion by amylase, tripsin and lipase
Describe what happens in the small intestine	Absorption of water and digested food Chemical digestion by enzymes Protease - protein to amino acids Sucrase - sucrose to glucose+fructose Maltase - Maltose to glucose
How is the small intestine adapted for the absorption of nutrients?	1.Villi & microvilli increase surface area 2.Thin lining (only one cell thick) so short exchange pathway to the blood and lymph 3.Good blood supply 4.Lacteals help transport fats and fatty acids 5.Mitochondria for active transport
What does bile do?	1. Neutralises chyme so create optimum conditions for enzymes in the small intestines 2. Emulsifies fats (breaks down large drops into smaller droplets, increasing the surface area for lipase to work to break down fats into fatty acids and glycerol. It allows faster digestion of fats)
What is the function of the gall bladder?	Stores bile
How are fatty acids and glycerol transported?	In the lymph
What happens in the large intestine (colon)?	Any remaining water is absorbed from the food

What causes tooth decay?	Food remains on teeth Bacteria use sugars to carry out respiration to produce acid that breaks down the enamel. The dentine is exposed and dentine dissolves more rapidly than enamel.
How can tooth decay be avoided?	Regular brushing with a good toothbrush, avoiding sugary foods and visiting the dentist regularly
What is plaque?	A layer that sticks to teeth and gums
State the function of fluoride ions	Strengthen tooth enamel'
Outline the arguments for and against the addition of fluoride to public drinking water [4]	For: * Fluoride strengthens tooth enamel * Cheap to supply Against: * Allergies/side effects * Bad taste
State the function of salivary glands	Secrete saliva that contains amylase to break down starch to glucose
State the function of rectum	Stores faeces
State the function of anus	Egestion
State the function of lacteal	Fat absorption
State the function of blood capillaries of the villus	Transport glucose and amino acids away
Draw a labelled diagram of a villus	
What is cholera?	A water-borne disease, caused by a bacterium
How does Vibrio cholera cause diarrhoea?	Releases toxins into the walls of the small intestines, causing chloride ions to pass out of the cells and into the intestinal lumen. This creates a low water potential inside the intestinal lumen, and so water moves by osmosis into the small intestine, resulting in the loss of water and mineral ions
What are the symptoms of cholera?	Diarrhoea and dehydration
How can we treat the symptoms of cholera?	oral rehydration therapy (ORT) to provide water, sodium ions lost from the blood, glucose for energy, and other ions lost in diarrhoea.
Why does giving someone water not treat cholera?	Water does not replace the ions that are being lost
Suggest the dangers to health of severe diarrhoea if not treated for a long time	Water and ions lost Dehydration Decrease in volume of blood Increase blood concentration Poor absorption of nutrients Death

B8. TRANSPORT IN PLANTS QUIZ CARDS

Question	Answer
State the functions of the xylem	1. Transport water and mineral ions from roots up the stem to the leaves (one-direction) 2. Support the stem + keep it upright
State the functions of the phloem	Transport sucrose and amino acids from source to sink\
Identify the position of xylem and phloem in roots	
Identify the position of xylem and phloem in leaf	
Identify the position of xylem and phloem in stems	
State the function of root hair cells	Absorb water (by osmosis) and mineral ions (by active transport) from the soil
Describe how root hair cells are adapted to increase rate of absorption	Have a large surface area
Describe how water enters a plant	By osmosis across root hair cells from the higher water potential in the soil to the lower water potential in the root cortex (=DOWN a concentration gradient)
Describe how roots are adapted to absorb water	Many roots and root hair cells - increase surface area increases the rate of absorption of water and mineral ions Thin, permeable cell wall - for absorption of water and mineral ions
Describe how mineral ions enter a plant	By active transport. Mineral ions are transported against the concentration gradient, from a low concentration in the soil, to a high concentration in the root hair cells using carrier proteins and energy (ATP)
State the pathway of water through a plant	Soil --> root hair cell --> root cortex cells --> xylem --> spongy mesophyll cells
Outline an investigation to investigate the pathway of water through a plant	Grow in coloured water / with a stain and then look at sections of the plant under a microscope 1) Add dye to water in a test tube 2) Put a celery into the test tube 3) Leave for a few minutes 4) Cut the celery and observe --> dye is in xylem (as water is transported)

How is water transported from the roots to the leaves?	Through xylem vessels
Define transpiration	loss of water vapour from plant leaves by evaporation of water at the surfaces of the mesophyll cells followed by diffusion of water vapour through the stomata
Explain why wilting occurs	Rate of transpiration > rate of absorption of water Water loss > water uptake
Describe the consequences of wilting	Less cell sap, which decreases turgor pressure causing plant cells to become flaccid so the leaves droop
Why might wilting be sometimes beneficial to a plant?	It reduces the surface area of the leaves, and the leaves face downwards, so there is a smaller surface area in direct light, reducing temperatures, and therefore reducing the rate of water loss (transpiration)
Explain how water moves through a plant	Transpiration occurs; the evaporation of water vapour through the stomata of the leaves by diffusion There is cohesion between the water molecules and adhesion between the water molecules and the side of xylem, so water is pulled up the xylem in a continuous column This creates a transpiration stream This lowers the water potential of the xylem in the roots, causing water to move from the higher water potential in the roots to the lower water potential of the xylem by osmosis This in turn lowers the water potential of the root hair cells, causing water to move from the higher water potential in the soil to the lower water potential of the root hair cells by osmosis
Define cohesion	Water molecules attracted to other water molecules
Define adhesion	Water molecules sticking to the sides of the xylem vessel
What is the transpiration stream?	The continuous flow of water from the roots to the leaves, created from transpiration, cohesion and tension
Outline an investigation to find out the effects of temperature and humidity on transpiration rate	Use a potometer. Set up the same species of plant in different environmental conditions. Set up a control. Measure the rate of water uptake over several hours/days
What does a potometer measure?	The rate of water uptake (from which we can infer this is similar to the rate of transpiration - but it won't be exactly the same as some water will have been used for photosynthesis!)
When are stomata usually open?	During the day when light is available for photosynthesis, the plant will also require carbon dioxide from the atmosphere (however, plants will close their stomata if in hot, dry conditions causing excessive transpiration)
Name 4 factors that affect the rate of transpiration	Temperature, humidity, light intensity, wind speed
Describe the effect of light intensity on transpiration	As light intensity increases, the rate of transpiration increases up to a maximum (that is determined by other conditions e.g. humidity/temperature)
What is humidity?	A measure of the quantity of water vapour in the air
Describe the effect of humidity on transpiration	As humidity increases, the rate of transpiration decreases
Describe the effect of temperature on transpiration	As temperature increases, the rate of transpiration increases
Describe the effect of wind speed on transpiration	As wind speed increases, the rate of transpiration increases
Explain the effect of light intensity on transpiration	Higher light intensity, more stomata open, so more diffusion of water vapour
Explain the effect of humidity on transpiration	Higher humidity, more water vapour in the air, reducing the concentration gradient for water vapour, so there is little diffusion
Explain the effect of temperature on transpiration	Higher temperature, faster evaporation of water as the water molecules have more kinetic energy
Explain the effect of wind speed on transpiration	Higher wind speed, faster movement of water vapour in the atmosphere away from the surface of the leaf, reducing the concentration gradient for water vapour, so there is little diffusion
Define translocation	The movement of sucrose and amino acids in phloem by active transport: – from regions of production (source) – to regions of storage OR to regions where they are used in respiration or growth (sink)
What is a source of sucrose in a plant?	Regions of production - the leaves

What is a sink of sucros in a plant?	Regions of storage, respiration and/or growth
Describe the structure of xylem vessels	columns of dead, empty cells with thick walls and no end walls
Describe the structure of phloem vessels	living cells that contains some cytoplasm and mitochondria
Describe how the xylem is adapted for its function [6]	Xylem transport water from the roots to the leaves for photosynthesis. Xylem vessels are dead and hollow Have lignin to provide support to the stem No end walls Walls are waterproof so water doesnt flow out and creates a continuous column of water called the transpiration stream

B9. TRANSPORT IN ANIMALS QUIZ CARDS

Question	Answer
Define circulatory system	a system of blood vessels with a pump and valves to ensure one-way flow of blood
Define single circulation	Blood flow through the heart once every complete circuit (e.g. fish)
Define double circulation	Blood flows through the heart twice every complete circuit (e.g. in humans). It keeps oxygenated and deoxygenated blood separated. There is one system between the heart and lungs, and one between the heart and body
What are the advantages of double circulation?	The left ventricle contracts with more force to create greater pressure for oxygenated blood to be delivered to the rest of the body. Oxygenated and deoxygenated blood do not mix
Which blood vessel carries blood away from the heart?	arteries
Which blood vessel carries blood under high pressure?	arteries
Compare the circulatory system of fish to animals	Fish have a single circulatory system (blood flows through the heart once during a cycle), whereas animals have a double circulatory system (blood flows between the heart and lungs, and the heart and body). In fish blood is oxygenated at the gills, whereas in animals this occurs at the lungs Both fish and animals transport blood in arteries and veins
What is the name of the valve in the pulmonary artery?	semi-lunar valve
What is the part of the heart that separates left from the right?	septum
Draw and label the structures of heart	① deoxygenated blood from body vena cava ⑤/⑪ semi-lunar valves ↳ prevent backflow of blood ② right atrium contracts ③ atrioventricular valves ↳ stop backflow of blood ④ right ventricle contracts Pulmonary = Lungs Aorta ⑩ oxygenated blood AWAY to body Pulmonary artery ⑥ deoxygenated blood AWAY from heart TO lungs Pulmonary vein ⑦ oxygenated blood AWAY from lungs TO heart Left atrium contracts ⑧ Atria-ventricular valves ③ ↳ prevents backflow of blood Left ventricle contracts ⑩ Septum ⑫ ↳ separate oxygenated & deoxygenated blood Arteries = AWAY Veins = TOWARDS / IN
Name the blood vessel that carries deoxygenated blood from the right ventricle	pulmonary artery (pulmonary=lungs, artery=away from the heart)
Name the blood vessel that carries oxygenated blood away from the heart	aorta (artery=away. Oxygenated=to the body)
What does the pulmonary vein do?	Carry oxygenated blood from the lungs to the heart
Name the type of blood vessel that carries blood to the heart	vein
Name the blood vessel that carries deoxygenated blood under low pressure	Vena cava (low pressure=vein. Deoxygenated=from the body)
What is the function of atrioventricular valves?	Atrioventricular valves close when the ventricle contracts so prevent backflow of blood from the ventricle to the atrium
What is the function of semi-lunar valves in the aorta?	Close when blood is in the aorta (=higher pressure), preventing backflow from the aorta into the left ventricle
Name the valves that separate the atrium from the ventricle	atrioventricular valves

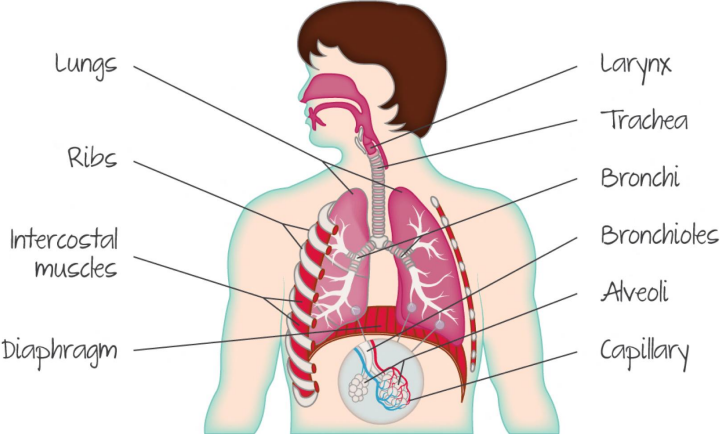
What is the function of the atrioventricular valves?	Prevent backflow of blood from the ventricles into the atria when the ventricles contract
Name the valves in the aorta and pulmonary artery	semi-lunar valves
Outline the function of the semi-lunar valves	Prevent backflow of the blood from the arteries into the ventricles when the ventricles contract
What is the function of the septum?	Separates the two sides of the heart and therefore prevents deoxygenated blood on the right side of the heart mixing with oxygenated blood on the left side of the heart
Why do the atria have less muscular walls than the ventricles?	The atria only need to contract to force the blood into the ventricles a short distance away, whereas the ventricles need to contract and force blood out of the heart
Where in the heart is the thickest wall?	left ventricle
Why is the left ventricle wall thicker than the right?	More muscular tissue in the left ventricle wall, in order to create a contraction to create more pressure to push blood around the body, whereas the right ventricle only needs to create enough pressure to push blood to the lungs
List the order of the structures blood passes through the heart if starting in the right atrium	right atrium (atrioventricular valves), right ventricle, (semi-lunar valves) pulmonary artery, lungs, pulmonary vein, left atrium, (atrioventricular valves) left ventricle, aorta, (semi-lunar valves), body, vena cava
Describe how blood travels from the lungs to the right ventricle	Oxygenated blood travels from the lungs along the pulmonary vein into the left atrium. The left atrium contracts, causing the atrioventricular valves to open and blood to enter the left ventricle. The atrioventricular valves close and the left ventricle contracts, causing the semi-lunar valves to open and blood to flow along the aorta towards the rest of the body to deliver oxygen. The semi-lunar valves close to prevent backflow of blood from the aorta into the left ventricle
What does an ECG do?	Monitors the activity of the heart by looking at the electrical activity of the heart
What creates the heart sounds?	closing of the valves
How can you monitor the heart?	ECG, Measure pulse, listen to valves closing
Name the blood vessel that supplies blood to the heart	coronary artery
why does the heart need its own blood supply?	To supply oxygen, for the cardiac muscles to carry out aerobic respiration, to release energy for muscles to contract.
What effect does exercise have on pulse rate? Why?	Pulse rate increases. So that the heart pumps more blood to the respiring cells, supplying more oxygen for aerobic respiration, to release energy for muscles to contract
If you investigate the effect of different types of exercise on pulse rate what is the IV, DV, CV?	IV - Different types of exercise DV - Pulse rate before, during / after CV - length of time of exercise, age/gender of participants etc.
List risk factors associated with coronary heart disease	smoking, stress, high salt diet, high fat diet, age, gender
How can someone reduce the risk of coronary heart disease	Don't smoke, low salt diet, regular exercise
How can coronary heart disease be treated?	aspirin to thin the blood, surgery (stents, angioplasty and heart by-pass)
Describe the structure of an artery	thick muscular and elastic walls, no valves (except aorta and pulmonary artery)
Explain how the structure of an artery is related to its function	thick muscular and elastic walls to withstand high blood pressure, and to stretch and recoil to maintain blood pressure, no valves as blood is under high pressure
Describe the structure of a vein	semi-lunar valves present, wider lumen, less muscular and elastic walls than arteries
Explain how the structure of a vein is suited to its function	semi-lunar valves to prevent backflow, wider lumen to reduce friction maintaining blood flow, less muscular and elastic walls as blood under low pressure
What is the function of a capillary?	Exchange of substances between blood and cells
Describe the structure of a capillary	Very narrow lumen, walls that are one cell thick, pores/holes in the capillary wall

Explain how the structure of a capillary is suited to its function	Capillaries have a thin lining that have walls that are one cell thick = short diffusion distance = increased rate of diffusion Small and narrow capillaries = blood moves slowly for exchange of substances between blood and tissue fluid Pores/holes in capillary wall = allows the movement of small molecules
Which arteries break the rule, and do have valves in?	aorta and pulmonary artery
What is the function of arterioles?	Carry blood from arteries to capillaries. Receive nerve impulses and change diameter (links to temperature control in homeostasis topic 14)
What is the function of venules?	Carry blood from capillaries to veins
What is the function of shunt vessels?	links an artery directly to a vein, bypassing capillaries
What is tissue fluid?	some parts of blood plasma that leave the blood and bathe cells
What is lymph?	When tissue fluid enters lymph capillaries
What is lymph made up of?	white blood cells, but no red blood cells or proteins
How is lymph moved around the body?	muscles surrounding the lymphatic vessels contract and move lymph Valves in the lymph vessels prevent backflow
Name 4 components of blood	red blood cells, white blood cells, platelets and plasma
What is the function of red blood cells?	contains haemoglobin which combines reversibly with oxygen to form oxyhaemoglobin to transport oxygen around the body
What is the function of the two types of white blood cells?	phagocytes carry out phagocytosis, and lymphocytes produce antibodies
What is the function of platelets?	blood clotting
What is the function of plasma?	transport of blood cells, ions, nutrients, hormones and carbon dioxide
How do blood clots form?	Platelets stimulate blood clotting Soluble fibrinogen is converted to insoluble fibrin Using thrombin enzyme To form a fibrin mesh that traps red blood cells and prevent pathogens from entering Hardens to form a scab
Why is it important that blood clots?	Prevents pathogens entering, prevents blood loss

B10. DISEASES AND IMMUNITY QUIZ CARDS

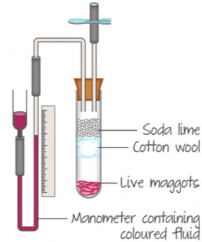
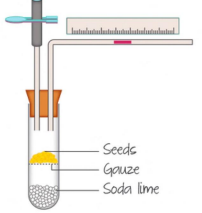
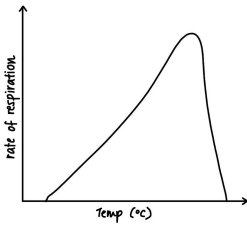
Question	Answer
Define pathogen	a disease-causing organism
Define transmissible disease	a disease in which the pathogen can be passed from one host to another
Define active immunity	defence against a pathogen by antibody production in the body
Define passive immunity	short-term defence against pathogens using antibodies from another individual
How might pathogens be transmitted?	direct contact (e.g. blood or body fluids), or indirectly (e.g. contaminated food, from air, animals)
How can we control the spread of disease?	Good hygiene, waste disposal, hygienic food preparation, sewage treatment
List the body defences against diseases	mechanical barriers (skin and hairs in the nose) chemical barriers (mucus and stomach acid) cells (phagocytosis and antibody production)
How do pathogens make you feel ill?	bacteria and viruses produce toxins, viruses cause cell damage/kill host cells
What is the function of the two types of white blood cells?	phagocytes carry out phagocytosis lymphocytes produce antibodies and antitoxins
What is an antigen?	a protein with a specific shape that induces an immune response
What is phagocytosis?	Phagocytes ingest and digest pathogens using enzymes
What do antibodies do?	antibodies can attach to it and stop the pathogen moving. Some antibodies kill pathogens directly by piercing holes in cell walls, causing water to enter and the cell to burst
Describe the immune response when a pathogen infects the body	Lymphocytes release antibodies, that attach to antigens destroying the pathogen. Phagocytes ingest and digest the pathogen (phagocytosis). Memory cells form
What does a vaccine contain?	A harmless/dead/weakened pathogen
Describe how a vaccine makes you immune	A harmless /dead/weakened pathogen is injected. The foreign antigens are detected by lymphocytes. Lymphocytes produce antibodies. Memory cells are formed that continue to produce the correct antibody. If the pathogen infects the person, memory cells produce a high concentration of antibodies very quickly (=immunity as the pathogen is destroyed before it makes someone feel ill)
Explain the role of vaccination in controlling the spread of disease	If a high proportion of the population is vaccinated, the pathogen cannot spread. This is herd immunity.
Name a disease caused by the immune system targetting and destroying its body cells	Type 1 Diabetes caused by the immune system destroying cells in the pancreas so not enough insulin is made
Explain what is meant by passive immunity	short-term defence against pathogens; no immune response/immediate protection/no memory cells produced/ no antibodies produced by the body ; antibodies acquired from elsewhere (not from own body) e.g. across placenta/breast-feeding/ breast milk
Describe how a baby acquires passive immunity	Antibodies cross the placenta and are in breast milk from the mother
Describe the differences in appearance of phagocytes, lymphocytes and red blood cells [3]	Lymphocytes have larger nucleus Phagocytes are irregular C-shaped cells Red blood cells have a biconcave shape
Describe the symptoms and treatment of type 1 diabetes [5]	Fatigue Thirst Dizzy Inject Insulin Controlled diet

B11. GAS EXCHANGE IN HUMANS QUIZ CARDS

Question	Answer
List the features of gas exchange surfaces in humans?	large surface area, thin surface, good blood supply and good ventilation with air
Describe and explain how the lungs are adapted for gas exchange	Large number of alveoli so <u>very large surface area</u> Alveoli have thin walls so <u>short distance for diffusion</u> Moist surface so <u>oxygen dissolve</u> before diffusion Many capillaries provide a good blood supply so oxygen supplied quickly & carbon dioxide removed quickly which <u>maintains a steep concentration gradient</u> Good ventilation with air which <u>maintains steep concentration gradient</u>
Draw and label the gass exchange system in humans	 The diagram illustrates the human respiratory system. On the left, labels point to the Lungs, Ribs, Intercostal muscles, and Diaphragm. On the right, labels point to the Larynx, Trachea, Bronchi, Bronchioles, Alveoli, and Capillary. The lungs are shown in pink, the trachea and bronchi in white, and the diaphragm and intercostal muscles in red.
Define ventilation	The movement of air between the environment and the lungs via inhalation and exhalation
Describe the functions of the cartilage	Cartilage is flexible yet has more rigidity than other tissues. It serves to keep the trachea open. Allows flexibility Form incomplete rings around trachea/bronchi Protect trachea Reduces resistance to movement of air
Describe the functions of the intercostal muscles, ribs and diaphragm	Produces volume and pressure changes in the thorax and leads to ventilations
Describe and explain how exercise affects breathing rate	An increase in physical activity leads to an increase in respiration. An increase in respiration leads to an increase in carbon dioxide in the blood. This increase in carbon dioxide is detected by the medulla in the brain. The brain sends nerve impulses to the external intercostal muscles and diaphragm to contract to cause more regular and deeper breaths. THEREFORE --> increases rate and depth of breathing
Explain why the heart rate increases when we exercise	Adrenaline stimulates an increase in heart rate Increase in carbon dioxide concentration/acidity detected in the blood More muscle contraction Muscles require more energy Increase in the cellular demand for oxygen Rate of aerobic respiration increases More blood containing oxygen supplied to muscles More blood containing carbon dioxide supplied to the lungs

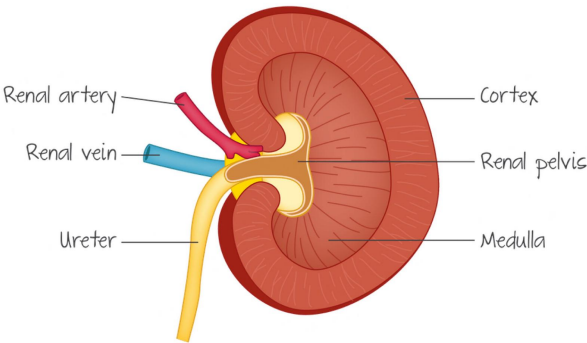
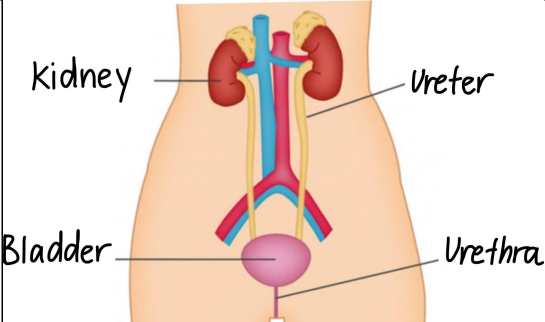
Plan an investigation to test the effect of exercise on breathing rate	IV: Doing exercise and not doing exercise. E.g. sit still versus 30 star jumps or sit still versus running 20 stairs DV: Count the number of breaths in a set period of time e.g. 1 minute Method: 1) Measure resting breathing rate 2) Exercise 3) Measure breathing rate every 2 minutes 4) Repeat for 5 trials
Describe the test for carbon dioxide	Bubble gas / exhale through limewater. If it turns from clear to cloudy if carbon dioxide is present
Compare the composition of inhaled and exhaled air	Inhaled air: more oxygen, less carbon dioxide, less water vapour Inhaled & exhaled air: same nitrogen, same noble gases
Compare and explain the composition of oxygen in inhaled and exhaled air	More oxygen in inhaled air than exhaled as oxygen diffuses into the blood as a reactant for cellular aerobic respiration
Compare and explain the composition of carbon dioxide in inhaled and exhaled air	More carbon dioxide in exhaled air than inhaled as carbon dioxide diffuses out of the blood as a waste product of cellular aerobic respiration
Compare and explain the composition of water vapour in inhaled and exhaled air	More water vapour in exhaled air than inhaled as water vapour diffuses out of the blood as a waste product of cellular aerobic respiration
Describe the process of inhalation	The diaphragm contracts & flattens. The external intercostal muscles contract (and the internal intercostal muscles relax). The causes the ribs to move up and out. The volume in the thorax increases. This decreases the pressure in the thorax Since atmospheric pressure (outside the lungs) is greater, air moves into the lungs.
Describe the process of exhalation	The diaphragm relaxes & curves upwards (becomes dome-shaped – due to pressure from the organs beneath – liver, stomach) The internal intercostal muscles contract (and the external intercostal muscles relax) The causes the ribs to move down and in The volume in the thorax decreases. This increases the pressure in the thorax Since atmospheric pressure (outside the lungs) is lower than in the thorax, air moves out of the lungs.
Describe the role of goblet cells and cilia in the airways	Goblet cells secrete mucus. Pathogens and dust that enter the trachea get trapped in the mucus. The cilia wave to move the mucus, dust and pathogens away from the lungs. At the back of the through the mucus is swallowed.

B12. RESPIRATION QUIZ CARDS

Question	Answer
List uses of energy	Muscle contraction Protein synthesis for making enzymes and antibodies Passage of nerve impulses Maintenance of a constant core body temperature Absorption of nutrients in the gut by active transport Making new cell membranes during growth
Does respiration require enzymes?	Yes - respiration always involves the action of enzymes in cells
Define aerobic respiration	Chemical reaction in cells that uses oxygen to break down nutrient molecules to release energy
State the word equation for aerobic respiration	oxygen + glucose → carbon dioxide + water + energy
State the balanced symbol equation for aerobic respiration	$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$
Plan an investigation that investigates the uptake of oxygen on respiring organisms	1. Germinating seeds or small arthropods such as woodlice are placed in one test tube of a respirometer. 2. Potassium hydroxide underneath the organism captures carbon dioxide produced in respiration. 3. The more the organism respire the more oxygen it uses. This change in gas volume causes a change of pressure in the capillary tube between the test tubes. 4. The level of the dye in the capillary tube alters according to the amount of respiration. 5. Measure the change in the volume of the dye 6. Repeat the investigation and calculate a mean Ensure the arthropods are not harmed in any way and are fed/provided with an environment that replicated their natural living conditions. Do not allow the arthropods to come into contact with the potassium hydroxide. Potassium hydroxide is corrosive, so wear gloves and goggles at all times, use the lowest volume and concentration possible.  ① measure position of fluid in manometer ② leave for 1 hr ③ measure final reading ↑ respiration = ↑ O₂ used = ↑ Δ gas volume = Δ pressure = Δ reading
Plan an investigation that investigates the effect of temperature on rate of respiration	 ① measure initial reading ② place in water bath (10°C, 20°C...50°C) ③ leave for 1 day ④ measure final reading ↑ respiration = ↑ CO₂ produced = ↑ Δ reading  1. Put germinating seeds in test tubes into water baths at different temperatures for 30 minutes (e.g. not in a water bath, 10°C, 30°C, 50°C, 70°C) 2. Attach an oxygen probe connected to a data logger to the top of each test tube 2. Collect data over 24 hours 3. Repeat and calculate a mean To stay safe, ensure water baths are not set above 70°C to minimise the risk of scalding, wear goggles to prevent hot water entering the eyes.

Define anaerobic respiration	the chemical reactions in cells that break down nutrient molecules to release energy without using oxygen
State the word equation for anaerobic respiration in muscles	glucose → lactic acid + a little energy
State the word equation for anaerobic respiration in yeast	glucose → carbon dioxide + alcohol + a little energy
State the balanced symbol equation for anaerobic respiration in yeast	$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
When does respiration take place?	All of the time
Where does aerobic respiration occur?	mitochondria
Compare when photosynthesis and respiration take place	Photosynthesis takes place during the daytime, respiration takes place all of the time
How are muscle cells different to a typical animal cell?	Have more mitochondria
Why do muscles cells have many mitochondria?	In order to carry out more aerobic respiration in order to release energy, for the muscles to contract
Name equipment that could be used to measure respiration in living organisms	An oxygen probe and data logger, or carbon dioxide probe and data logger OR a respirometer
If a carbon dioxide probe was inserted into a sealed container with germinating seeds, what do you predict the data will show? And why?	Carbon dioxide levels increase, as aerobically or anaerobically respiring plants react glucose with oxygen to produce carbon dioxide, water and energy
Where does anaerobic respiration occur?	cytoplasm
Compare the energy released from aerobic and anaerobic respiration	More energy is released per glucose molecule from aerobic respiration than anaerobic respiration
When does anaerobic respiration occur in humans?	Short, vigorous exercise
What is the effect of anaerobic respiration on the human body?	Oxygen debt & a build-up of lactic acid can cause muscle cramps and fatigue
How is oxygen debt removed during recovery?	1. Aerobic respiration of lactic acid in the liver 2. Continuation of fast heart rate to transport lactic acid in the blood from the muscles to the liver 3. Continuation of deeper and faster breathing to supply oxygen for aerobic respiration of lactic acid
Define oxygen debt	Amount of oxygen needed to break down lactic acid produced in anaerobic respiration
Why is there oxygen debt during exercise?	Muscles respire anaerobically Lactic acid produced Demand for oxygen increases Increases rate of respiration Limited supply of oxygen to muscle tissue Heart rate/breathing rate not increased fast enough
Explain why the athlete has a high ventilation rate after the exercise has finished [5]	During the exercise, anaerobic respiration occurred so there is a build up of lactic acid and oxygen debt so more oxygen is required . There is a high concentration of carbon dioxide in the blood from aerobic respiration , which is detected by the medulla in the brain , which sends impulses to the external intercostal muscles and diaphragm to contract more and stimulate a high ventilation rate, increasing the depth of breathing too.

B13. EXCRETION QUIZ CARDS

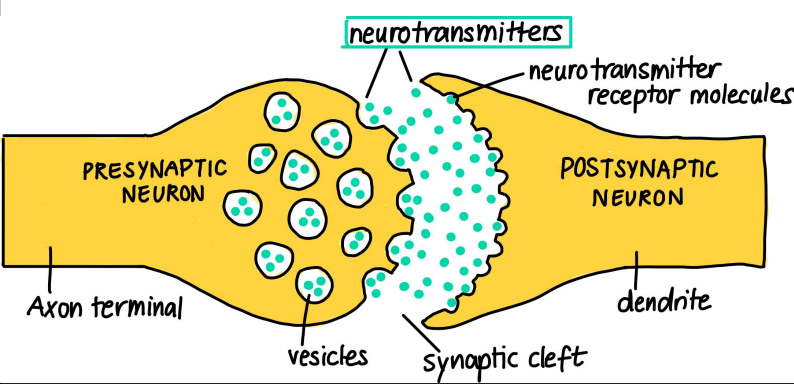
Question	Answer
List the two main waste products	Carbon dioxide and urea
Distinguish between excretion and egestion	Excretion is the removal of waste chemicals made in tissues during metabolism from the body. Egestion is the removal of waste from the digestion of food (i.e. faeces)
What is carbon dioxide a waste product of, why is it excreted and how is it excreted by?	CO ₂ is a waste product of aerobic cellular respiration Excreted because it is toxic in high concentrations to the body Excreted by exhalation in the lungs
What is urea a waste product of, why is it excreted and how is it excreted by?	Urea is a waste product from excess amino acids Excreted because it is toxic in high concentrations to the body Excreted by the kidney in urine
Where is urea formed?	Liver
Describe the formation of urea	Deamination (removal of nitrogen-containing part of amino acids to form urea) occurs in the liver One amino acid is converted into fat or carbohydrate, the other molecule is ammonia which combines with carbon dioxide to form urea
List the functions of the liver	Stores glucose as glycogen Assimilation of amino acids to make proteins (include plasma proteins like fibrinogen) Breaks down excess amino acids into urea Converts fatty acids and glycerol into fat which can be stored Produces cholesterol from fats Produces bile to emulsify fats Breaks down toxins such as alcohol
Describe what happens to amino acids in liver cells [3]	Excess amino acids form urea Deamination occurs, which is the removal of the nitrogen-containing group to form ammonia Amino acids are joined by peptide bonds to form proteins .
Define deamination	removal of nitrogen-containing part of amino acids to form urea
Draw a labelled diagram of a kidney	
Draw a labelled diagram of the urinary system	
Name the blood vessel that carries blood to the kidneys	renal artery
Name the blood vessel that carries blood away from the kidneys	renal vein
Name the tube that carries urine from the kidney to the bladder	Ureter

What is the name of the tiny tubes inside the kidney?	kidney tubules
What does the urinary system consist of?	kidneys, ureters, bladder and urethra
What is the function of the kidneys?	Filter the blood Excrete urea and excess water and salts
Describe how the kidneys filter the blood	Glucose, amino acids, water and mineral ions are filtered out of the blood in the glomerulus by diffusion into the bowman's capsule of the kidney. Proteins are too large so they do not diffuse through into the bowman's capsule All of the glucose and amino acids are reabsorbed by active transport Some of water and mineral ions are selectively reabsorbed by active transport . Urea, excess water and excess mineral ions forms urine which is transported along the ureters to be stored in the bladder and excreted through the urethra
Describe what happens to urine after leaving the kidney tubule [2]	Transported down the ureter Stored in bladder Excreted through the urethra
State the composition of urine	Urea, excess water and excess mineral ions
Explain why urine tests are a good indicator of how much salt has been consumed [2]	Excess salts excreted Some salts reabsorbed in kidney tubule People cannot remember how much they eat
If someone drinks a lot of water, what will the kidneys do?	Selectively reabsorb less water back into the blood. There will be a large volume of dilute urine
If someone sweats a lot during exercise, what will the kidneys do?	Selectively reabsorb more water back into the blood. There will be a small volume of concentrated urine
How is water lost from the body?	Urine, sweat, tears
What is the glomerulus?	A group of capillaries that allow exchange of materials in the bowman's capsule of the kidney
Describe the structure of a glomerulus [2]	A ball of capillaries that are one cell thick Have small pores Tangled network of capillaries
Why are proteins not filtered out of the blood in the bowman's capsule of the kidney?	Proteins are too large to pass through the partially permeable membrane
How is the kidney adapted for its function?	Microvilli to increase surface area for absorption, many mitochondria to release energy for active transport
Describe the factors that affect the volume and concentration of urine	Water intake - more water intake = greater volume of dilute urine Temperature - higher temperature = sweat more = lower volume of urine Exercise - more exercise = sweat more = lower volume of urine
What is dialysis?	Dialysis is the filtering of the blood to remove waste products when the kidneys are not functioning
What is a common indicator of kidney damage / failure?	If glucose is present in the urine
If a patient's kidneys fail, what options do they have?	Kidney dialysis or a kidney transplant
Describe how dialysis works	The dialysis fluid contains same concentration of glucose and salts as the patient's body , so there will be no NET movement of glucose and salts out of the blood . Dialysis fluid does NOT contain urea , so urea diffuses from a high concentration in the blood into an area of lower concentration in the dialysis fluid, through the dialysis membrane , and leaves the patient's body .
What is a dialysis membrane?	A membrane that allows small molecules to pass through the pores but prevents large molecules from pass through (i.e. mimicking the cell membrane in the glomerulus)
Why is there no net movement of glucose during dialysis?	The dialysis fluid contains the same concentration of glucose as the patient's blood
Will glucose molecules move between the patients blood and the dialysate (the dialysis fluid)?	Yes - There just wont be any net movement. All molecules always move

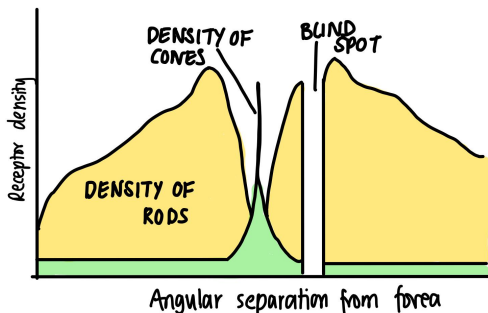
What is a kidney transplant?	Replacing the diseased kidneys with a healthy kidney
What are the disadvantages of a kidney transplant?	Shortage of organ donors There may not be a suitable donor There is a risk of tissue rejection (immune system destroys the kidney as recognises it as non-self. The antigens determine the tissue type) Risk of complications during surgery Immunosuppressant drugs must be taken for the rest of their life to reduce the chance of tissue rejection but this can lead to the patient becoming ill from minor infections
What are the advantages of a kidney transplant?	Greater quality of life after successful kidney transplant Do not need to go for dialysis (=time consuming) anymore = saves time Less diet restrictions
What are the disadvantages of kidney dialysis?	Regular hospital visits (2-3 times a week) Dialysis takes a long time to be carried out (several hours) More costly in the long run Only works for limited time Discomfort and reduced quality of life Patients have to eat a restricted diet e.g. less protein to reduce excess amino acids and reduce the build-up of toxic urea in the blood in between dialysis treatments
What are the advantages of a kidney dialysis?	Energy is not needed as molecules diffuse Prevent surgery

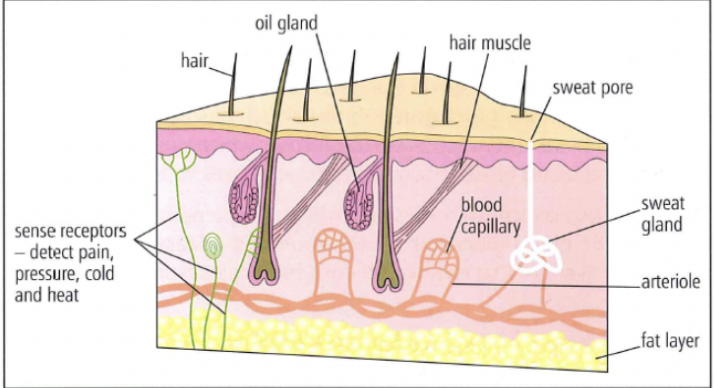
B14. COORDINATION AND CONTROL QUIZ CARDS

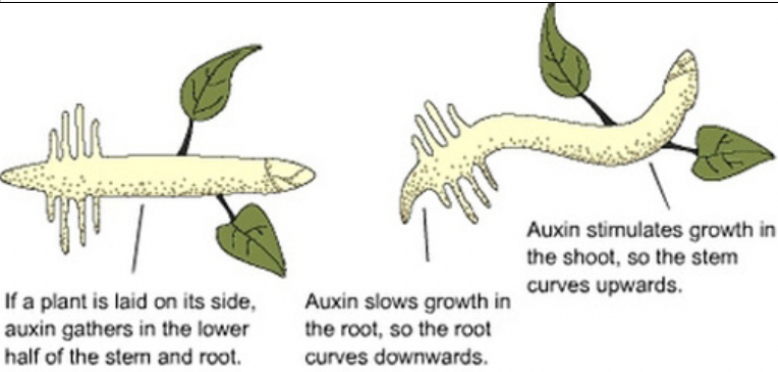
Question	Answer		
Define nerve impulse	an electrical signal that passes along nerve cells called neurones in one direction		
What is the central nervous sytem made up of?	Brain and spinal cord		
What is the peripheral nervous system made up of?	Cranial and spinal nerves It connects the central nervous system to the body		
What are cranial nerves?	Connect the brain with all other organs in head, thorax, abdomen		
What are spinal nerves?	Connect the brain with the arms, thorax, abdomen and legs		
What is the human nervous system made up of?	central nervous system (brain and spinal cord) and peripheral nervous system		
What is nerve made up of?	Lots of neurones / nerve cells		
Define voluntary action	Reactions that we choose to make (involve the brain) e.g. catching a ball		
Define involuntary action	A automatic response that does not involve a decision/thought		
Define reflex action	Automatically and rapidly intergrating and coordinating stimuli with the responses of effectors (mscules and glands)		
Compare and contrast voluntary and involuntary actions		Voluntary actions	Involuntary actions
	Similarities	✓ stimulus , receptors ✓ sensory & motor neurones ✓ effector & response	
	Differences	✓ sensory neurones - CNS Slower	✓ Relay neuron - bypass brain Quicker
Draw a sensory neuron			
Draw a relay neuron			
Draw a motor neuron			
Describe how a structure of a neurone is related to its function [3]	Long = transmit impulse over long distances Mitochondria to release energy for transmission impulse Vesicles release neurotransmitters Receptors on post-synaptic neurone and vesicles on pres-synaptic neurone causes the impulse to flow in one direction		
Describe the process of a voluntary action	Seven Rainy Summers Can Make Everyone Remember: Stimulus, receptor, sensory neuron, central nervous system, motor neuron, effector, response		

A person kicks a football. Describe the voluntary response	The stimulus is the ball coming towards the person. The receptors in the eyes detect the stimulus. A nervous impulse travels down a sensory neuron When the impulse arrives at a synapse (gap between neurones), the vesicles in the pre-synaptic neuron release a neurotransmitter that diffuses across the synapse, and reaches the post-synaptic neuron and binds with receptor molecules. The nervous impulse reaches the brain and spinal cord (central nervous system), which coordinates the response The nervous impulse then travels down a motor neuron, to the effector, the muscles in the leg. The response is the person kicks a football
Describe the process of an involuntary action	Seven Rainy Summers Can Make Everyone Remember: Stimulus, receptor, sensory neuron, RELAY NEURON , motor neuron, effector, response
Suggest why reflexes occur in people who are unconscious [1]	Involuntary Receptors still function
A person touches a hot object. Describe the reflex response	The stimulus is the hot object touching the skin. The receptors in the skin detect the stimulus. A nervous impulse travels down a sensory neuron When the impulse arrives at a synapse (gap between neurones), the vesicles in the pre-synaptic neuron release a neurotransmitter that diffuses across the synapse, and reaches the post-synaptic neuron and binds with receptor molecules. The nervous impulse travels down a relay neuron (in the spinal cord, by-passing the brain) The nervous impulse then travels down a motor neuron, to the effector, the muscles in arm. The response is the hand moves away from the hot object
Suggest the advantages of having reflexes. You may refer to an example to illustrate your answer. [3]	Rapid response can protect us from danger An example is automatically moving our hand away when close to a hot object
What is the coordinator of a voluntary action?	The brain (i.e. it coordinates the response!)
What is the role of human nervous system,?	coordinated and regulate body functions
What is a stimulus?	A change in the environment
Define synapse	Junction between two neurones. Ensures the impulse travel in one direction only
What are receptors?	Cells that detect a stimulus
Where in the body are receptors?	skin, ears, eyes, tongue, nose
What are effectors?	Muscles or glands that carry out the response to the stimulus
Why are neurones surrounded by myelin?	It is a layer rich in fat, that provides insulation, and therefore the impulses can travel very quickly
Draw and label a synapse	
Describe what happens at the synapse	When the impulse arrives at a synapse (gap between neurones), the vesicles in the pre-synaptic neuron release a neurotransmitter that diffuses across the synapse, reaches the post-synaptic neuron and binds with receptor molecules. The nervous impulse then continues down the neuron.
How can drugs affect synapses?	Influencing the release of neurotransmitters or interacting with the receptors on the post-synaptic membrane

What affect do excitatory drugs have on synapses?	stimlate the release of neurotransmitters, increasing activity
Name an example of an excitatory drug	amphetamines - increase brain activity / alertness
What affect do inhibitory drugs have on synapses?	reduce the effect of neurotransmitters / block the receptors that neurotransmitters bind it
Name an example of an inhibitory drug	heroin, beta-blockers. Reduce the feeling of pain
Define sense organs	group of receptor cells responding to specific stimuli (light, sound, touch, temperature, chemical)
Draw and label the human eye	
What is the function of the cornea?	refracts light
What is the function of the iris?	controls how much light enters pupil
What is the function of the lens?	focuses light onto retina
What is the function of the retina?	contains light receptors (cones and rods), some sensitive to light of different colours
What is the function of the optic nerve?	carries impulses to the brain
Describe the pupil reflex in bright light	To make the pupil diameter smaller the radial muscles relax and the circular muscles contract (Remember- muscles get short and fat when they are contracting)
Describe the pupil reflex in dim light	To make the pupil diameter larger the radial muscles contract and the circular muscles relax (Remember- muscles get short and fat when they are contracting)
Explain why the change in pupil size as the light is switched on is necessary [2]	So less light enters the eye to protect the retina/rods/cones
What type of response is a pupil reflex?	Involuntary response
Why are the radial and circular muscles of the eye antagonistic pairs?	work together in pairs - when one contracts, the other relaxes
What is accomodation?	The way the lens changes shape to focus at objects near and far away from the eye
How is light refracted so that it focuses on the retina?	Light is refracted by the corena and aqueous humour. The lens then focuses the light further.
Explain how light is focused to be able to see a distant object	The ciliary muscle relaxes (hole gets wider) The suspensory ligaments get tighter The lens gets flatter/elliptical (thin) shape Light is refracted less to focus on the retina
Explain how light is focused to be able to see a close object	The ciliary muscle contracts (hole gets smaller) The suspensory ligaments get slacker The lens gets rounder/spherical (fatter) Light is refracted more to focus on the retina
What is the function of rods?	Detect light of low intensity and convert the light to electrical impulses. Rods provide night vision and give black and white vision . They are sensitive in dim light
What is the function of cones?	Sensitive to high light intensity, three different kinds of cones absorbing light of different colours for colour vision Each cone is connected to its own neuron to the brain so gives a very detailed image

Outline how humans are able to see in colour [3]	Cones detect light of three different wavelengths and converts light into electrical impulses which is carried along the optic nerve to the brain where it interprets the impulse in terms of colours .									
Does the fovea contain rods or cones?	Lots of cones - most sensitive part of the retina									
Describe the distribution of rods and cones on the retina	No rods and cones on fovea (centre of retina). Many rods, few cones on the rest of the retina 									
Define a hormone	a chemical substance, produced by a gland and carried by the blood, which alters the activity of one or more specific target organs									
What hormones does the adrenal gland secrete?	adrenaline - allows 'flight or fight response' by increasing heart rate, breathing rate, widening pupils									
What hormone does the pancreas secrete? And what is the function?	insulin/glucagon - controls blood sugar levels									
What hormone do the testes secrete? And what is the function?	testosterone - stimulates the secondary male features and body hair									
What hormone do the ovaries secrete?	oestrogen									
Describe the effects of adrenaline	Increases heart rate - more oxygen to muscle cells, more respiration, faster contraction to run away Increases breathing rate - more oxygen uptake and carbon dioxide excreted Widens pupils Increases blood glucose concentration Fight or flight response									
Describe situations when adrenaline would be released from the gland into the blood [3]	During a fight or flight response When in dangerous situations Display aggression									
Describe the effects of testosterone	Increase muscle mass, increase body hair, controls growth and development									
Describe the effects of oestrogen	Growth of breasts, more body hair, menstrual cycle, controls growth and development									
Compare and contrast the nervous and endocrine system	<table><tr><td></td><td>NERVOUS SYSTEM</td><td>ENDOCRINE SYSTEM</td></tr><tr><td>Similarities</td><td colspan="2"> • occurs in humans • involved in survival </td></tr><tr><td>Differences</td><td> • Rapid • short lived • Travel along neurons • small target area • wired system </td><td> • slower • longer lived • Travel in the blood • large target area - whole organ • wireless system </td></tr></table>		NERVOUS SYSTEM	ENDOCRINE SYSTEM	Similarities	• occurs in humans • involved in survival		Differences	• Rapid • short lived • Travel along neurons • small target area • wired system	• slower • longer lived • Travel in the blood • large target area - whole organ • wireless system
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Differences	• Rapid • short lived • Travel along neurons • small target area • wired system	• slower • longer lived • Travel in the blood • large target area - whole organ • wireless system								
Define homeostasis	The maintenance of a constant internal environment. The control of internal conditions within set limits									
Define negative feedback	A regulatory mechanism that returns a condition to its optimum e.g. body temperature to 37C									
Give examples of constant internal conditions in the human body	Blood sugar (glucose) concentration. Water, carbon dioxide, oxygen concentration of the blood. pH of the blood. Body temperature									
Name 2 hormones that control blood sugar levels	insulin and glucagon									
What is glycogen?	A complex carbohydrate / polysaccharide made up of many glucose molecules. Stored in the liver and muscle cells									

Describe the response of the human body when blood glucose level decreases	Increase in blood glucose concentration is detected by the pancreas . The pancreas secretes GLUCAGON which travels in the bloodstream to the liver. The liver breaks down glycogen into glucose using enzymes in liver cells , which diffuses into the bloodstream , increasing blood glucose levels. This is an example of homeostasis/negative feedback .
Suggest why blood sugar levels may decrease	Long time since last meal / been asleep / after prolonged exercise
Describe the response of the human body when blood glucose level increases	Increase in blood glucose concentration is detected by the pancreas . The pancreas secretes INSULIN which travels in the bloodstream to the liver. The liver absorbs more glucose from the bloodstream and stores it as glycogen (this also occurs in the muscles) using enzymes in liver cells , decreasing blood glucose levels. This is an example of homeostasis/negative feedback .
Suggest why blood sugar levels may increase	Recently eaten a meal
Describe how negative feedback ensures blood glucose concentration is controlled	The pancreas makes insulin to stimulate the liver to absorb glucose, decreasing blood glucose concentration, or makes glucagon to stimulate the liver to break down glycogen into glucose, increasing blood glucose concentration
Name the largest organ in the body	The skin
List 3 functions of the skin	Protects the body from damage, stops pathogens from entering, prevents too much water loss, detects changes in temperature, detects pressure (touch) and pain, loses heat by conduction, convection, radiation and evaporation
Identify on a diagram: hairs, hair erector muscles, sweat glands, receptors, sensory neurones, blood vessels and fatty tissue	 Figure 11.2.1 The structure of human skin.
List the responses that occur in the human body if body temperature increases	Sweating may occur (and a smaller volume of urine is produced). Evaporation of sweat causes cooling of the body. Arterioles (blood vessels supplying the skin capillaries) dilate, more heat is lost by radiation from the surface of the skin, hair erector muscles relax (causing hairs to stay flat on the skin's surface) Behavioural responses (taking off outer layers of clothing)
Describe what happens when body temperature increases	The thermoregulatory centre in the brain detects changes in blood temperature, causing sweating to occur, as evaporation of sweat cools the body. Arterioles dilate (vasodilation) supplying more blood to the surface of the skin, and more heat loss by radiation.
List the responses that occur in the human body if body temperature decreases	Arterioles (blood vessels supplying the skin capillaries) constrict, so less heat is lost by radiation from the surface of the skin. Sweating does not occur. Muscles contract causing shivering requires aerobic respiration, which releases heat energy. Hair erector muscles contract, causing hairs to stand up trapping a layer of insulating air. Behavioural responses (putting on more clothes)
Describe what happens when body temperature decreases	The thermoregulatory centre in the brain detects changes in blood temperature, causing shivering to occur; the contraction of muscles requires aerobic respiration which releases heat energy. Hair erector muscles also contract causing hairs to stand on end trapping a layer of insulating air. Arterioles constrict (vasoconstriction) supplying less blood to the surface of the skin, and less heat loss by radiation.
Describe how the hair, receptors, sweat glands, and fat help to maintain a constant body temperature [4]	The hair contracts and stands up when too cold and as air is a good insulator, it will prevent heat loss The temperature receptor detects the change in temp and sends an impulse to the CNS The sweat gland produces sweat, which evaporates to cool down the body The fat acts as an insulator.

Explain how negative feedback is involved in the control of body temperature [3]	The change in temperature is detected and the body carries out a corrective action to keep the temperature at the norm .
Explain the importance of regulating body temperatures in humans [4]	Prevent enzymes from denaturing Optimum temperature for enzymes Maintain constant rate of reaction Prevent heat stroke/chills
Explain how vasodilation and vasoconstriction are involved in maintaining body temperature	When body temperature decreases, arterioles (blood vessels supplying the skin capillaries) constrict (vasoconstriction) supplying less blood to the surface of the skin, and less heat loss by radiation. When body temperature decreases, arterioles dilate (vasodilation) supplying more blood to the surface of the skin, and more heat loss by radiation.
What causes type 1 diabetes?	Occurs when the body cannot produce insulin
What are the symptoms of type 1 diabetes?	Feeling thirsty Tired/fatigue Weight loss Glucose present in urine Vomitting
What are the treatment for type 1 diabetes?	Injecting insulin Regular glucose tests Controlled diet - avoid sugary foods
Define gravitropism	a response in which parts of a plant grow towards or away from gravity
Define phototropism	a response in which parts of a plant grow towards or away from the direction from which light is coming
Define positive phototropism	a response in which parts of a plant grow towards the direction from which light is coming e.g. shoots!
Define negative gravitropism	a response in which parts of a plant grow away from gravity e.g. the shoots!
Describe how shoots bend towards the light	Auxin is made in the tip of the shoot. There is an unequal distribution of auxin. Auxin accumulates on the shaded side of the shoot. Auxin causes cell elongation, so the shoot bends towards the light
Describe how roots bend towards gravity	There is an unequal distribution of auxin. Auxin accumulates on the LOWER side of the root. Auxin inhibits cell elongation, so the root bends downwards towards gravity
What is the difference between the effect of auxin on shoots and roots?	In shoots, auxin STIMULATES cell elongation. In roots, auxin INHIBITS (i.e. stops) cell elongation
What is a function of synthetic auxin?	Weedkillers - absorbed by broad leaved weeds = increases growth rate, the plant cannot photosynthesise/produce enough glucose fast enough for the rapid growth rate, so the plant dies
Draw a diagram of a plant shoot and root and the distribution of auxin in response to gravity	 If a plant is laid on its side, auxin gathers in the lower half of the stem and root. Auxin stimulates growth in the shoot, so the stem curves upwards. Auxin slows growth in the root, so the root curves downwards.
Explain why it is important for their early growth that the roots and shoots of the seedlings respond to gravitropism [2]	So the roots grow downwards to absorb more water and the shoots grow upwards so that the leaves can absorb more sunlight for photosynthesis
Suggest how weedkillers spread throughout the plant [2]	Weedkillers spread by diffusion and are absorbed where it enters the phloem and are translocated .

B15. DRUGS QUIZ CARDS

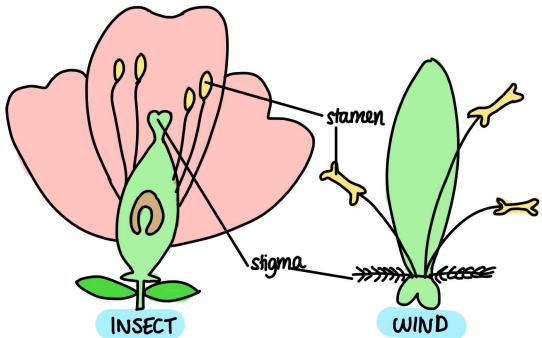
Question	Answer
Define a drug	Any substance taken into the body that modifies or affects chemical reactions in the body
What do antibiotics kill?	Bacteria
Do antibiotics kill viruses?	No
Why don't antibiotics kill viruses?	Viruses are not cells e.g. do not have cell walls that are targetted by antibiotics. Viruses are INSIDE host cells, so to target a virus the host cell would also have to be targetted
Describe how antibiotics are used in the treatment of bacterial infection	Antibiotics can inhibit bacterial reproduction or kill bacterial cells by targetting parts of the cell e.g. creating holes in cell wall, or preventing cell wall formation, leading to leakage of the cells contents
How do bacteria reproduce?	binary fission
What are antibiotic resistant bacteria?	Bacteria that are not killed by antibiotics
List strategies to reduce the spread of antibiotic resistant bacteria e.g. MRSA	Only take antibiotics when necessary , ensure patients always complete the course of antibiotics so all bacteria are killed, do not use antibiotics in livestock feed
Suggest how the problem of antibiotic resistance can be limited [4]	Dont overuse antibiotics and only prescribe less often Always finish a course of the antibiotic treatment Rotate use of antibiotics --> not the same each time Dont use for viral/fungal infections Practise good hygiene to prevent spread of infections Develop new antibiotics
Suggest why are people given the advice to finish a course of antibiotics [2]	To kill all the bacteria so prevent the bacteria from becoming resistant
Describe how bacteria become antibiotic-resistant bacteria	1) Mutation leading to variation (some resistant and some non-resistant) 2) Overproduction of offspring leading to competition for food and space ---> struggle for survival 3) Antibiotics kill bacteria that are non-resistant 4) Bacteria that are resistant survive to reproduce (by binary fission) and pass on their advantageous alleles to offspring 5) Process of natural selection 6) Over time, whole population becomes resistant to the antibiotic
Why do antibiotic resistant bacteria form?	Overuse of antibiotics, patients not finishing the full course, poor infection control in hospital and clinics
What is penicillin	The first antibiotic discovered by Alexander Fleming
What is MRSA?	<i>Methicillin-resistant Staphylococcus aureus</i> - resistant to many antibiotics
Explain the possible effect of the X-rays on the bacteria [3]	X-rays can increase the rate of mutation = change in DNA = increase the chance of developing antibiotic-resistant bacteria
Suggest why some antibiotics are used less frequently than others [3]	The bacteria may be resistant to some antibiotics due to natural selection and overuse . Some antibiotics are too expensive and some have side-effects
Explain why antibiotics cannot be used to control HIV [2]	HIV are viruses and antibiotics are not effective against viruses because they do not have a cell wall and are located inside host cells so antibiotics cannot get to them
Explain how heroin affects the nervous system, limited to its effect on the function of synapses	Heroin has a chemical structure similar to endorphins in the brain. Endorphins provide release from pain and stress. Heroin binds to endorphin-receptor sites on the post-synaptic membrane, preventing the neurotransmitter from binding, and preventing the nerve transmission

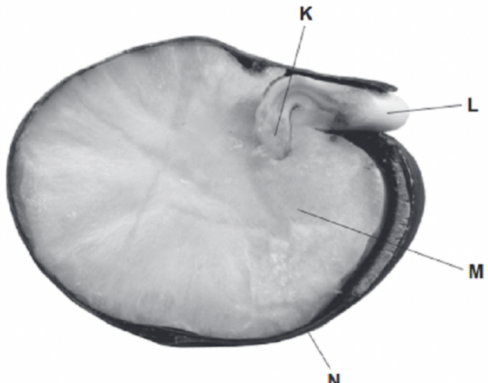
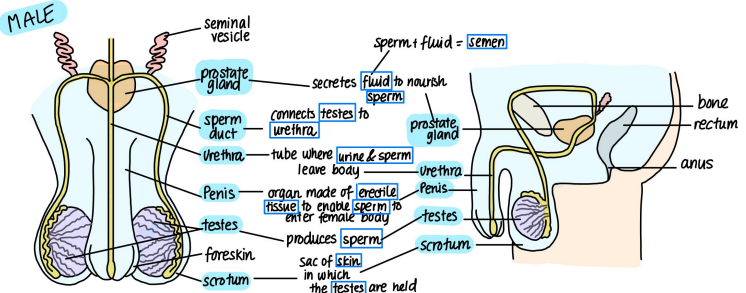
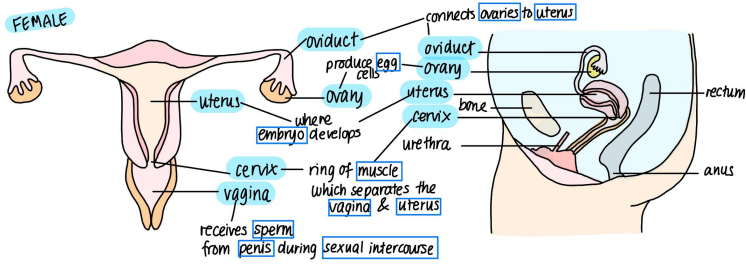
Describe the effects of heroin consumption and excessive alcohol consumption	Effect	Impact
	Powerful depressants	Both alcohol and heroin are depressants despite both drugs giving feelings of wellbeing when initially taken. This feeling of depression can lead to individuals continuing to abuse the drug.
	Reaction times	Reaction times increase (speed of reactions decrease) whilst individuals have the drug in their system. This means that they are not safe to operate machinery, including driving.
	Self control	Inhibitions are lost when individuals are under the influence. This means that they can make unreasonable decisions which they wouldn't if they were not under the influence of the drug.
	Addiction	Both drugs are very addictive. When addiction takes over the drug is all that matters to the individual and carrying on with a 'normal' life can be difficult.
	Withdrawal	When the user stops taking the drug withdrawal symptoms can begin which are difficult to manage. Withdrawal symptoms include craving, diarrhoea and vomiting, sweating, twitching, mood swings and crying.
	Social implications	Often addicts cannot continue with work because the drug has taken over their normal life. Heroin and alcohol can be costly so addiction often increases crime.
List some risks of injecting heroin?	Injections can cause the spread of HIV (human immuno-deficiency virus), addiction, withdrawal symptoms, speed of reaction decreases making it dangerous to operate machinery/drive, lack of self-control, increased crime levels as addicts need money to pay for their addiction	
Which part of the body does heroin effect?	Synapses in the brain	
What can happen due to excessive alcohol consumption?	Alcohol is a recreation drug and depressant. Excessive alcohol consumption can lead to liver damage (the liver is the site of breakdown of alcohol and other toxins)	
Describe the effects of nicotine in tobacco	Is a stimulant = It is rapidly absorbed across the alveoli and into the blood stream. It interacts at the synapse binding as a stimulant, thus increasing heart rate. In addition it narrows arterioles, increasing blood pressure and causes platelets to become more sticky increasing likelihood of blood clots	
Describe the effects of tar in tobacco	Tar, a sticky black substance, cannot pass across the alveoli walls but instead becomes stuck to their surface. It irritates all linings of the airways and increases mucus production. This mucus accumulates because cilia, which are involved in removing the mucus, are also damaged. This leads to 'smoker's cough'.	
Describe the effect on the gas exchange system of tar [2]	Carcinogens in tar can lead to lung cancer Tar sticks to the walls of the alveoli Produces more mucus	
Describe the effects of carbon monoxide in tobacco	A poisonous gas which is more readily taken up by red blood cells than oxygen. It therefore reduces the volume of oxygen in the blood (as much as 10%) this puts added strain on the heart which may already be beating harder and faster due to the nicotine. Carbon monoxide binds to haemoglobin permanently	
Describe the effect on the gas exchange system of carbon monoxide [2]	Binds to haemoglobin Reduces volume of oxygen transported to respiring cells Increases heart rate	
What is the function of the liver?	breakdown of toxins	
Describe the effects of excessive alcohol consumption on the liver [2]	Fatty liver and cirrhosis and liver failure	
Where is alcohol broken down?	Liver. (But when consumed in excess alcohol damages the liver - cirrhosis of the liver)	
What is an anabolic steroid and what effect does it have?	Similar to the male hormone testosterone, mimicks the protein-building effects increasing muscle growth, improving an athletes strength and endurance	
List some harmful side-effects of taking anabolic steroids in a. men b. women	A. In men, increased aggression, impotence, baldness, kidney and liver damage, development of breasts. B. In women, facial and body hair, irregular periods	
Name diseases caused by smoking	chronic obstructive pulmonary disease (COPD), lung cancer, coronary heart disease	

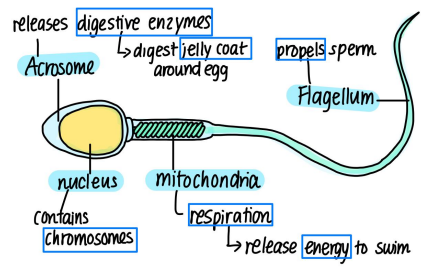
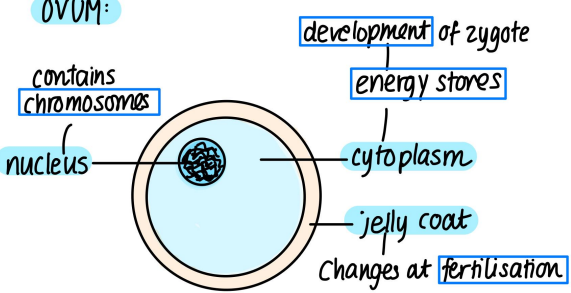
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B16. REPRODUCTION QUIZ CARDS

Question	Answer
Define asexual reproduction	a process resulting in the production of genetically identical offspring from one parent
What are the advantages of asexual reproduction in crop production?	Higher yield, Higher economic gains Farmers can keep desired characteristics
What are the disadvantages of asexual reproduction in crop production?	Prone to whole crop population being killed at once, Reduced diversity of native species, human manipulation required which is costly
What are the advantages of asexual reproduction for wild species?	1. Rapid Populating - large quantities of offspring made. 2. No Mates Required - Finding a mate can be very difficult. 3. No True Investment as asexual reproducers do not have to carry their offspring for a long amount of time and produce more than one at a time. This makes it a quick and inexpensive process for them in the terms of time.
What are the disadvantages of asexual reproduction for wild species?	1. No Diversity: Since the traits of only one parent is passed on, all of the offspring are genetically identical. 2. Prone To Extinction: All of the same traits also means all of the same weaknesses. Parasites and other predators that have evolved to kill just one of the organisms can take out the entire population.
Give examples of species who reproduce asexually	Plants, bacteria, yeast, starfish
Define sexual reproduction	a process involving the fusion of the nuclei of two gametes (sex cells) to form a zygote and the production of offspring that are genetically different from each other
Define fertilisation	the fusion of gamete nuclei
What controls the activities of the cell?	Nucleus
What does haploid and diploid mean?	haploid is a nucleus that contains half the number of chromosomes (n) of a diploid nucleus. A diploid nucleus has a full set of chromosomes/chromosomes in pairs (2n)
Is the nucleus of a gamete haploid or diploid?	Haploid
Is the nucleus of a zygote haploid or diploid?	Diploid
What are the advantages of sexual reproduction in crop production?	Genetic variation because offsprings inherit alleles from different plants Allows mutations to be expressed More able to adapt to changing environments More natural selection occur
What are the disadvantages of sexual reproduction in crop production?	Lower yield, Less economic gains, Offspring phenotype (characteristics) cannot be controlled
What are the advantages of sexual reproduction for wild species?	Meiosis leads to variation as the offspring inherit alleles from different parents = more genetic variation = more able to adapt to changes in the environment = reduced risk of extinction . Allows mutations to be expressed and natural selection can occur. Less competition among offsprings
What are the disadvantages of sexual reproduction for wild species?	Time consuming, Requires energy and therefore plentiful resources, Need to find a mate
Draw and label the structures of a flower	
What is the carpel made up of?	stigma, style and ovary - female reproductive part
What is the stamen made up of?	Anther and filament - male reproductive part
State the function of the stigma	Sticky surface to trap pollen, through which pollen tubes grow

State the function of the ovary	Contains ovules that develop into seeds when fertilised																											
State the function of the petals	Brightly coloured to attract insects																											
State the function of the sepals	Protects the growing flower																											
State the function of the anthers	Produce plant gametes (pollen)																											
Distinguish between an insect pollinated and wind pollinated flower	<table><tr><th></th><th>Insect Pollinated</th><th>Wind Pollinated</th></tr><tr><td>Petals</td><td>Large, brightly coloured – to attract insects</td><td>Small, dull in colour – no need to attract insects</td></tr><tr><td>Smell</td><td>Sweet smell – to attract insects</td><td>No scent – no need to attract insects</td></tr><tr><td>Nectar</td><td>Contains nectar – to attract insects</td><td>No nectar – no need to attract insects</td></tr><tr><td>Pollen quantity</td><td>Not much required – less wastage than with wind pollination</td><td>Huge quantities required – most of the pollen doesn't reach another flower</td></tr><tr><td>Pollen characteristic</td><td>Sticky or spiky – to stick to insects</td><td>Light, dry and smooth – so it doesn't clump together and can be blown by the wind</td></tr><tr><td>Anthers position</td><td>Firm and inside flower – to brush against insects</td><td>Loose and outside flower – to release pollen into the wind</td></tr><tr><td>Stigma position</td><td>Inside flower – so that insect brushes against it</td><td>Outside flower – to catch the drifting pollen</td></tr><tr><td>Stigma characteristic</td><td>Sticky – so that pollen from insect sticks to it</td><td>Stick but also feathery and / or net like – to catch drifting pollen</td></tr></table>		Insect Pollinated	Wind Pollinated	Petals	Large, brightly coloured – to attract insects	Small, dull in colour – no need to attract insects	Smell	Sweet smell – to attract insects	No scent – no need to attract insects	Nectar	Contains nectar – to attract insects	No nectar – no need to attract insects	Pollen quantity	Not much required – less wastage than with wind pollination	Huge quantities required – most of the pollen doesn't reach another flower	Pollen characteristic	Sticky or spiky – to stick to insects	Light, dry and smooth – so it doesn't clump together and can be blown by the wind	Anthers position	Firm and inside flower – to brush against insects	Loose and outside flower – to release pollen into the wind	Stigma position	Inside flower – so that insect brushes against it	Outside flower – to catch the drifting pollen	Stigma characteristic	Sticky – so that pollen from insect sticks to it	Stick but also feathery and / or net like – to catch drifting pollen
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Draw a diagram of an insect pollinated and wind pollinated flower																												
Define pollination	the transfer of pollen grains from the anther to the stigma																											
Define self-pollination	the transfer of pollen grains from the anther of a flower to the stigma of the same flower or different flower on the same plant																											
Define cross-pollination	transfer of pollen grains from the anther of a flower to the stigma of a flower on a different plant of the same species																											
When does fertilisation occur in a plant?	when a pollen nucleus fuses with a nucleus in an ovule																											
What are the implications to a species of cross-pollinating?	More variation in the population so more likely to be able to adapt to changing environmental conditions by natural selection/not all killed by the same pathogen. Reliance on pollinators																											
What are the implications to a species of self-pollinating?	No reliance on pollinators but less genetic variation and therefore less capacity to respond to changing environmental conditions by natural selection																											
Describe the events that occur in flowering plants after fertilisation to form a seed [4]	A zygote forms and divides by mitosis to grow into an embryo . The ovule forms the seed and the radicle/cotyledon/testa forms																											
Describe what happens after pollination that results in fertilisation in flowering plants [3]	Pollen tube grows from pollen grain and reaches the ovule through the micropyle. The male gamete travels down the pollen tube and it fuses with the female gamete to form a zygote.																											
Define germination	When the seed splits and grows into a plant																											
List the conditions required for germination	water, oxygen, suitable temperature																											
Label K to N	K - Plumule																											

	L - Radicle M - Cotyledon N - Testa
Describe the growth of the pollen tube and its entry into the ovule followed by fertilisation	Pollen lands on the stigma (pollination). From the pollen grain grows a pollen tube which reaches and then enters the ovule through the micropyle. The nucleus from the pollen grain travels through the tube to reach the egg in the ovule. When the pollen grain and egg nuclei fuse fertilisation has occurred
Describe seed formation	After fertilisation the zygote divides and grows into an embryo. The ovule forms the seed with the embryo inside it. The ovary forms the fruit with the seed inside it
Name the male and female sex organs	Testes & ovaries
Explain what is meant by the term ovulation [2]	The release of egg from the ovary
Identify the parts of male reproductive system	
Identify the parts of the female reproductive system	
State the function of the ovary	produces eggs, oestrogen and progesterone
State the function of the oviduct	carries the egg from the ovary to the uterus
State the function of the uterus	Location where the zygote implants, grows into an embryo and then into a foetus
State the function of the urethra	Carries urine (urea, excess water and mineral ions) out of the body
State the function of the cervix	ring of muscle at the opening of the uterus. Helps hold the growing foetus in place
What is the scientific name for sex cells?	Gametes
What is the male and female gamete?	sperm, eggs
State the function of the prostate gland	Secretes fluid to nourish the sperm (sperm+fluid = semen)
State the function of the sperm duct	Connects testes to urethra
State the function of the urethra	Tube where urine and sperm leaves the body
State the function of the penis	Organ made of erectile tissue to enable the sperm to enter the female body
State the function of the testes	Produces sperm
State the function of the scrotum	Sac of skin in which the testes are held

Compare the male and female gametes	Size - sperm are smaller than eggs. Movement - sperm have a flagellum to move, egg is moved by cilia in the oviduct. Energy store - sperm cells has a little sugar for respiration for energy for swimming, egg has protein&fat in the cytoplasm. Number of chromosomes - both have 23 chromosomes. Number produced - millions of sperm, one egg per month until menopause
Draw and label a sperm cell	SPERM CELL: 
Draw and label an egg cell	OVUM: 
Describe 3 ways the sperm is adapted for its function	Many mitochondria, to carry out respiration, to release energy to swim. Acrosome contains digestive enzymes to digest the jelly coat surrounding the egg, flagellum to swim towards to egg. (Flagellum is not always on mark schemes so make sure you know the other two aswell!)
Describe how the egg is adapted for its function	energy stores and a jelly coating that changes after fertilisation
Define fertilisation	The fusion of the nuclei from the male gamete and the female gamete
Name 1 hormone released by the testes	Testosterone
Name 2 hormones released by the ovaries	Oestrogen & progesterone
What happens during early development of a zygote?	Grows by mitosis into an embryo (ball of cells) that implants into the wall of the uterus
What is the function of the umbilical cord?	Attaches the fetus to the placenta. Contains blood vessels to transport blood to and from the placenta
What is the function of the placenta?	Site of exchange by diffusion - fetus to receive oxygen and nutrients, and remove carbon dioxide and waste (urea). Also provides a barrier to toxins and pathogens
List three functions of the placenta [3]	1. Protection against pathogens 2. Supplies oxygen to the fetus 3. Carries carbon dioxide away from the fetus (4. Barrier between blood systems)
Name some toxins & pathogens that can pass through the placenta and affect the fetus	nictotine, alcohol, HIV, rubella virus
If the placenta is a site of exchange, what adaptations does it have?	Large surface area, short distance for diffusion (there are 2 blood supplies - maternal blood and fetal blood - between which there is exchange of oxygen/nutrients/carbon dioxide/waste (urea))
What nutrients does the fetus receive from the mother via the placenta?	glucose, amino acids, fats, mineral ions, vitamins, water
What is the function of the amniotic sac?	Makes amniotic fluid
What is the function of the amniotic fluid?	Formed from filtered blood. Surround and protect the fetus e.g. when the mother makes sudden movements

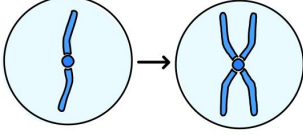
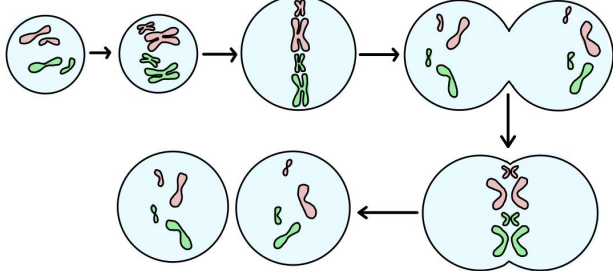
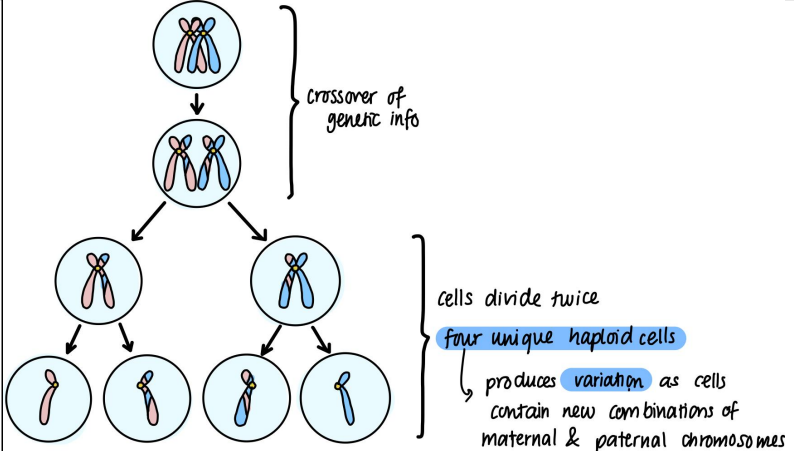
Describe the functions of the amniotic sac and amniotic fluid [4]	The amniotic sac contains amniotic fluid , which acts as a shock absorber to protect the fetus when the mother makes sudden movements . It creates a sterile environment and provides a constant temperature . Allows movement of fetus
What is gestation period?	The period of time between fertilisation and birth (in humans roughly 9 months)
Outline the growth and development of a fetus	Increases in complexity during early stages, and in size during later stages of pregnancy (Embryo then a fetus (has limbs and most organs) - then nerves and muscles - then body movements/fingernails/body hair)
Describe what happens during labour and birth	Amniotic sac breaks (also known as a woman's "waters breaking") and contractions of the muscles of the uterus wall, dilation (this means it gets wider) of the cervix, passage of the baby through the vagina, the umbilical cord is tied and cut (makes the belly button!), delivery of the afterbirth (this is when the placenta comes out of the vagina)
Describe the ante-natal care of the mother (NB. Ante means "before" Natal means "birth")	Special dietary needs (adequate quantities of calcium, iron, carbohydrate and protein), no smoking, no alcohol
List the advantages of breastfeeding	Passive immunity - antibodies from mother can be passed on to the baby Bond is formed between mum and baby Breast milk is free Milk is at the correct temperature Child less likely to develop diabetes Less risk of allergies
List the disadvantages of breastfeeding	Only the mother can feed infant determines the volume of milk consumed so not easy to monitor how much the baby is drinking medication and diet can affect breast milk may transfer viruses/HIV Embarrassment Painful nipples
List the advantages of bottle feeding formula	No social embarrassment of breast feeding in public, convenient that dad can feed the baby (not just mum) - although breast milk can be put into a bottle!
List the disadvantages of bottle feeding formula	Cost to buy formula, bottles may not be properly sterilised making the child ill, no passive immunity, no bond between mum and baby, more likely child develops childhood cancers and diabetes
What does testosterone in boys cause?	growth of male sex hormones, testes to make sperm, growth of facial hair, deepening of the voice, development of muscles
What does oestrogen in girls cause?	growth of female sex hormones, start of the menstrual cycle and periods, growth of body hair, growth of breasts, widening of hips
Name 4 hormones involved in the control of the menstrual cycle and where each is produced	FSH (pituitary gland), oestrogen (ovaries), progesterone (ovaries), LH (pituitary gland)
Describe the hormonal control of the menstrual cycle	The pituitary gland releases FSH (F = First hormone), causing the egg to mature into a follicle and stimulates the ovaries to release oestrogen. Oestrogen causes the lining of the womb/uterus to get thicker, inhibits the pituitary gland from producing FSH and stimulates the pituitary gland to secrete LH. LH causes the egg to be released from the ovaries. The ovaries produce oestrogen & progesterone which continues to thicken the lining of the uterus, inhibit FSH & LH.
Explain the role of FSH in the control of the menstrual cycle [3]	FSH stimulates follicle to grow and develop and stimulates the secretion of oestrogen
Describe the role of oestrogen in controlling the menstrual cycle [4]	It inhibits the release of FSH and stimulates the release of LH It causes the uterus lining to thicken and stimulates the repair of the uterus lining It stops the production of more eggs It prepares the uterus for implantation
Describe the roles of progesterone during the menstrual cycle and during pregnancy [3]	Progesterone thickens the uterus lining to prepare for implantation It maintains the uterus lining It inhibits LH Stops menstrual cycle in pregnancy

Describe the roles of oestrogen and progesterone during pregnancy [3]	They develop the lining of the uterus and prevent menstruation . They inhibit the release of FSH by the pituitary gland and prevent ovulation .		
How do FSH, oestrogen, progesterone and LH travel around the body	In the blood (they are hormones)		
Outline natural methods of birth control	Method	Details	
	Abstinence	Not having sexual intercourse.	
	Monitoring body temperature	Female body temperature peaks when ovulation occurs. Females can take their temperature everyday and avoid having intercourse when her temperature peaks.	
	Cervical mucus	Cervical secretions change before and during ovulation to assist sperm. Before and during ovulation secretions become clear and wet. Females would avoid having sexual intercourse while she observes these secretions.	
Outline chemical methods of birth control	Method	Details	Diagram
	IUD (intrauterine device)	A device made of copper, implanted into the uterus which makes it less likely that sperm survive and reduces the chance of implantation if an egg is fertilised.	
	IUS (intrauterine system)	Similar to the IUD but it releases hormones. These hormones make mucus more sticky which reduces the motility of sperm making it less likely they reach the egg. In some women it also stops ovulation in some females	
	Contraceptive pill	The pill releases oestrogen and progesterone which stops ovulation.	
	Implant	A small plastic tube that is inserted under the skin in the upper arm and it releases progesterone. This make mucus thicker preventing fertilisation and implantation and the likelihood of ovulation.	
	Injection	The injection administers progesterone which lasts for anywhere between 8-14 weeks depending on the type.	
Outline barrier methods of birth control	Method	Details	
	IUD (intrauterine device)	A device made of copper, implanted into the uterus which makes it less likely that sperm survive and reduces the chance of implantation if an egg is fertilised.	
	IUS (intrauterine system)	Similar to the IUD but it releases hormones. These hormones make mucus more sticky which reduces the motility of sperm making it less likely they reach the egg. In some women it also stops ovulation in some females	
	Contraceptive pill	The pill releases oestrogen and progesterone which stops ovulation.	
	Implant	A small plastic tube that is inserted under the skin in the upper arm and it releases progesterone. This make mucus thicker preventing fertilisation and implantation and the likelihood of ovulation.	
Outline surgical methods of birth control	Method	Details	
	Vasectomy	The sperm ducts are cut and tied stopping sperm being ejaculated. This can sometimes be reversed but is usually a method used by fathers who do not want anymore children.	
	Female sterilisation	This seals the oviducts stopping eggs from reaching the uterus and stopping sperm being able to reach the egg. Again this is mainly used by females who have a family and do not want anymore children.	
Outline how artificial insemination is carried out in humans [2]	Sperm is collected from donor and inserted into the uterus lining of the woman near the time of ovulation		

If a couple is unable to get pregnant, name 3 options they have to help them conceive	fertility drugs (Inject FSH early in the menstrual cycle that inhibits oestrogen and stimulates the development and maturation of the follicles so more eggs are released) artificial insemination (donor sperm inserted into female uterus near the time of ovulation and fertilisation occurs) invitro fertilisation (fertilisation takes place outside the body and the embryo is then implanted inside the mother's uterus)
Explain why the sperm must be placed in the uterus near the time of ovulation when carrying out AI? [3]	The sperm can only survive for a few days so must be inserted when the egg is released to increase the chance of fertilisation as fertilisation occurs in the oviduct
Explain why it is important that progesterone is secreted after ovulation [3]	To maintain the lining of the uterus for implantation and prevent the loss of the embryo through menstruation . Progesterone inhibits the release of LH .
Describe how FSH is used as a fertility drug and how it may allow a woman to conceive [2]	FSH is injected at the beginning of the cycle and it stimulates the development of follicles
Suggest why contraceptive pills do not contain FSH [3]	FSH stimulates the follicle to mature and develop and also stimulates the release of oestrogen FSH leads to ovulation FSH increases the chance of fertilisation
Explain why it is important that FSH is not secreted during pregnancy [2]	So no more eggs are released and no fertilisation occurs
Outline the stages of IVF	Extracting the woman's egg, the eggs are kept in a solution of oxygen and nutrients, fertilising the eggs with the male's sperm in a lab, implanting multiple embryos into the uterus (in the hope that one will develop!)
Discuss the social implications of contraception and fertility treatments	Some religions do not accept artificial methods of birth control. Some fertility treatments result in many eggs being fertilised but not all being given the opportunity to develop. Some are discarded or used for research. Contraception can limit population growth and some reduce the spread of infections. Some fertility treatments can lead to genetic screening which can have desirable and undesirable consequences depending on how it is regulated.
Define a sexually transmitted infection	An infection that is transmitted via body fluids through sexual contact
Name one STI	Human Immunodeficiency virus (HIV)
Describe how the spread of STIs can be controlled	Avoiding having unprotected sex is the only way STIs can be controlled.
Describe how HIV affects the immune system	HIV destroys lymphocytes so less lymphocytes = fewer antibodies produced = increased susceptibility to diseases = can lead to AIDS (Acquired immunodeficiency syndrome)
How is HIV transmitted?	unprotected sex, sharing needles, from mother to fetus, from blood products (e.g. blood transfusions in hospitals)
Describe what happens to viruses that leave white blood cells that are infected with HIV. [2]	The virus enters the blood and infects other white blood cells .
What is the difference between HIV and AIDS	HIV is the virus that infects the person, which can lead to AIDS (auto-immune deficiency syndrome)
Describe what is meant by the term development	Increase in complexity and tissues/organs form
Give one example of how insects are useful to farmers [1]	Pollination
Describe the role of stem cells in the growth of the embryo [2]	Stem cells are unspecialised cells that grow and divide by mitosis to become specialised cells when certain genes are expressed

B17. INHERITANCE QUIZ CARDS

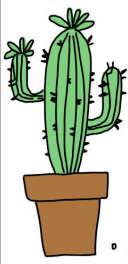
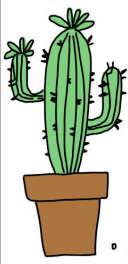
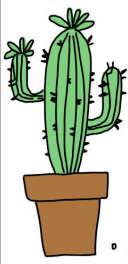
Question	Answer
Define inheritance	the transmission of genetic information from generation to generation
Define chromosome	thread-like structure of DNA, carrying genetic information in the form of genes
Define gene	length of DNA that codes for a protein
Define allele	a version of a gene
What chemical are genes made up of?	DNA (deoxyribose nucleic acid)
What sex chromosomes does a male have?	XY
What sex chromosomes does a female have?	XX
Describe the inheritance of sex in humans	The diagram illustrates the inheritance of sex in humans. At the top, a male (XY) is shown with arrows pointing to 'X' and 'Y' chromosomes. A female (XX) is shown with arrows pointing to two 'X' chromosomes. Below this, a Punnett square shows the possible combinations: XX (Female) and XY (Male). To the left of the Punnett square, a box contains the text: '1 : 1', 'XX : XY', '50% Female', and '50% Male'.
Define haploid nucleus	a nucleus containing a single set of unpaired chromosomes e.g. in gametes
Define diploid nucleus	a nucleus containing two sets of chromosomes (one inherited from the mother and one inherited from the father), e.g. in body cells
Are gametes haploid or diploid?	haploid
Are body cells haploid or diploid?	diploid - there is a pair of each type of chromosome
How many pairs of chromosomes are there in a human diploid cell?	23 pairs (46 chromosomes in total)
What are proteins made of?	amino acids
What determines the order of amino acids in a protein?	the sequence of bases in a gene is the genetic code for putting together amino acids in the correct order to make a specific protein
Where in a plant/animal cell is DNA found?	In chromosomes in the nucleus
How does DNA control cell function?	by controlling the production of proteins (some of which are enzymes), antibodies and receptors for neurotransmitters
If one strand of DNA is ATCCCG, what is on the complementary strand?	TAGGGGC (apples in trees - A binds to T. C binds to G)
Where in a cell does protein synthesis occur?	ribosomes
Describe how a protein is made	– the gene coding for the protein remains in the nucleus – mRNA molecules carry a copy of the gene to the cytoplasm – the mRNA passes through ribosomes – the ribosome assembles amino acids into protein molecules – the specific order of amino acids is determined by the sequence of bases in the mRNA
How do cells become specialised?	all body cells in an organism contain the same genes, but many genes in a particular cell are not expressed because the cell only makes the specific proteins it needs
Name 3 roles / functions of mitosis	in growth, repair of damaged tissues, replacement of cells and asexual reproduction
Define mitosis	nuclear division giving rise to genetically identical cells
Chromosomes duplicate BEFORE the start of mitosis. True or False?	True!

What are the products of mitosis?	Two genetically identical daughter cells (i.e. same number of chromosomes as the parent cell)
Explain how during mitosis the number of chromosomes is maintained between the parent cell and daughter cell	the copies of the chromosomes separate, maintaining the chromosome number (the parent cell and each daughter cell have the same number of chromosomes)
If a cell has 48 chromosomes, and undergoes mitosis, how many chromosomes will it have?	48
If a cell has 48 chromosomes, and undergoes meiosis, how many chromosomes will it have?	24 (=half 48!)
Draw a diagram to show mitosis occurring	Stages: - Before mitosis: <u>exact duplication</u> of chromosomes occur  - During mitosis: copies of chromosomes <u>separate</u>, maintaining the chromosome number 
Define stem cells	unspecialised cells that divide by mitosis to produce daughter cells that can become specialised for specific functions
Define meiosis	nuclear division that is reduction division in which the chromosome number is halved from diploid to haploid in cells
List differences between mitosis and meiosis	Mitosis - one division, same number of chromosomes in parent and daughter cells, daughter cells are genetically identical to parent cells, two daughter cells formed Meiosis - two divisions, half the number of chromosomes in daughter cell than parent cell, daughter cells genetically different to parent cells, four daughter cells formed
What is the function of meiosis	Produce gametes with half the number of chromosomes of the parent cell
Draw a diagram to show meiosis occurring	
How does meiosis produce variation?	Cells contain new combinations of maternal and paternal chromosomes
Define genotype	the genetic make-up of an organism in terms of the alleles present
Define phenotype	the observable features of an organism
Define homozygous	having two identical alleles of a particular gene

What is pure breeding?	two identical homozygous individuals that breed together
Define heterozygous	having two different alleles of a particular gene
If two heterozygous individuals are bred, is this pure-breeding?	No (two homozygous individuals only)
If one homozygous and one heterozygous individual is bred is this pure-breeding?	No (two homozygous individuals only)
Define dominant	an allele that is expressed if it is present
Define recessive	an allele that is only expressed when there is no dominant allele of the gene present
Explain how to use a test cross to identify an unknown genotype	A test cross allows you to work out the genotype of an unknown parent. You always cross the unknown parent with a homozygous recessive parent. By looking at the offspring we can determine the genotype of the unknown parent.
What does co-dominance mean?	If both alleles are present, then they are both expressed
Give an example of co-dominance	A & B in blood types
Explain what is meant by codominance and give an example [3]	When for two (or more) alleles , both alleles are expressed in a heterozygous individual. An example is AB blood group. Neither allele is dominant or recessive
If one parent is IAIO and one is IBIO, what is the chance the child will be blood group AB? Draw the punnett square, list the genotypes and phenotypes	Genotypes: 25% IAIB, 25% IAIO 25% IBIO, 25% IOIO Phenotypes: 25% AB, 25% A 25% B, 25% O Answer = 25% chance AB
If one parent is IAIO and one is IBIO, what is the chance the child will be blood group O? Draw the punnett square, list the genotypes and phenotypes	Genotypes: 25% IAIB, 25% IAIO 25% IBIO, 25% IOIO Phenotypes: 25% AB, 25% A 25% B, 25% O Answer = 25% chance O
Define sex-linked characteristic	a characteristic in which the gene responsible is located on a sex chromosome and that this makes it more common in one sex than in the other
Name an example of sex linkage	colour-blindness (the gene is located on the X chromosome, and because there is no homologous region on the Y chromosome it is more prevalent in males)
Why is it important for gametes to be haploid?	So chromosome number does not double at fertilisation So chromosome number remains constant from generation to generation
Discuss whether sickle cell trait is an example of codominance [2]	Yes, because both alleles are expressed and there are 3 phenotypes
Explain why it is important that meiosis occurs during the production of gametes. [2]	Meiosis produces genetically different haploid cells so it contains half the number of chromosomes and gives variation . This prevents the number of chromosomes from doubling each generation .
Explain why genotypes of the seeds are not all the same [2]	The gametes are different and produced by meiosis that gives rise to variation . The pollen grains are collected by different plants

B18. VARIATION AND SELECTION QUIZ CARDS

Question	Answer
Define variation	differences between individuals of the same species
Distinguish between phenotypic variation and genetic variation & give an example of each	Phenotypic variation is caused by both genetic and environmental factors e.g. height, weight Discontinuous variation is mostly caused by genes alone, e.g. A, B, AB and O blood groups in humans
What does continuous variation result in?	caused by the interaction between many genes and the environment. Creates a range of phenotypes between two extremes, e.g. height in humans
What does discontinuous variation result in?	caused by genes only. Creates a limited number of phenotypes with no intermediates, e.g. tongue rolling (i.e. either have the characteristic or you don't - there is no inbetween!)
Define mutation	genetic change
Define gene mutation	change in the base sequence of DNA
How are new alleles formed?	From random mutations!
Mutations always occur randomly, but what speeds up the mutation rate of cells / living organisms?	ionising radiation (x-rays, gamma rays) and some chemicals (e.g. chemicals in cigarette smoke)
Describe the symptoms of sickle-cell anaemia	Fatigue, Shortness of breath, Heart palpitations, Pain (because sickle shaped red blood cells have abnormal haemoglobin, and so do not transport oxygen around the body as effectively = less oxygen to respiring cells)
What causes sickle-cell anaemia?	One base change in the gene for haemoglobin! = abnormal haemoglobin = sickle-shaped red blood cells
State the genotypes of a homozygous sickle-cell anaemia patient and a homozygous healthy/normal patient	Remember we write this as the allele on the Haemoglobin gene (You will see the gene as Hb or H)
State the genotype of someone who is heterozygous for sickle-cell anaemia	They are a carrier of sickle cell anaemia
Explain why sickle cell anaemia is a form of discontinuous variation [3]	Genetic condition Distinct groups of phenotypes with no intermediates Either have it or don't
Why is sickle-cell anaemia still present in the population?	Being a carrier (heterozygous) for sickle cell anaemia (HbA HbS) provides a selective advantage. These people are resistant to malaria (disease transmitted by mosquitoes)
Explain how natural selection is responsible for the distribution of the allele for the abnormal form of haemoglobin (H ^A s) [5]	Areas with malaria = high % of sickle cell allele If homozygous recessive = reduced life span bc of sickle cell anaemia If homozygous dominant = susceptible to malaria If heterozygous = selective advantage bc resistant to malaria = survive and reproduce = pass on alleles to offspring 1/4 chance of homozygous recessive
Define adaptive feature	an inherited feature that helps an organism to survive and reproduce in its environment
Explain how the features of a polar bear help survival	Black skin - absorb solar radiation, keeping the polar bear warm sharp claws - catch prey thick layer of fat - for heat insulation etc
Define fitness	the probability of an organism surviving and reproducing in the environment in which it is found

Explain the adaptive features of hydrophytes to their environments	HINT: Hydro = water. Think of water lilies (plant that lives in water). Specialised leaf shape to create a large surface area for light absorption/photosynthesis Air-filled cavities in leaves and stems to provide an internal atmosphere of gases required for photosynthesis/respiration (as oxygen and carbon dioxide diffuse through the water much more slowly than through the air) Roots for anchorage Few roots as no need to absorb water Little/no waxy cuticle on leaves as do not need to conserve water/reduce transpiration Air spaces for buoyancy						
Explain the adaptive features of xerophytes to their environments	HINT: Think about a cactus! Spines/needles instead of leaves to reduce the surface area for transpiration Rolled leaves with stomata on the inside to prevent water loss by transpiration Sunken stomata to allow a layer of humidity to build up reducing water loss by evaporation Thick waxy cuticle on the epidermis to prevent water loss Hairs that trap a layer of moist air reducing transpiration Wide-spreading shallow roots & deep roots to maximise water absorption Vertical stem to avoid mid-day sun on a large surface area						
Compare and contrast hydrophytes and xerophytes	<table border="1"> <thead> <tr> <th>Hydrophytes</th><th>Xerophytes</th></tr> </thead> <tbody> <tr> <td> - plants that grow in or on water - e.g. water lilies </td><td> - plants that need very little water - e.g. cactus </td></tr> <tr> <td>  ↑SA = √ light absorption / photosynthesis - air-filled cavities in leaves & stem = internal atmosphere of gases (CO₂ & O₂) - roots = anchorage - few roots = no need to absorb water - little/no waxy cuticle = no need to conserve water </td><td>  - spines/needles = ↓ SA - rolled leaves + stomata on inside = ↓ water loss by transpiration - sunken stomata = build up of layer of humidity = ↓ water loss by transpiration - thick waxy cuticle = ↓ water loss - hairs = trap layer of moist air = ↓ transpiration - wide spreading shallow + deep roots = maximise water absorption - vertical stem = avoid sun on large SA </td></tr> </tbody> </table>	Hydrophytes	Xerophytes	- plants that grow in or on water - e.g. water lilies	- plants that need very little water - e.g. cactus	 ↑SA = √ light absorption / photosynthesis - air-filled cavities in leaves & stem = internal atmosphere of gases (CO₂ & O₂) - roots = anchorage - few roots = no need to absorb water - little/no waxy cuticle = no need to conserve water	 - spines/needles = ↓ SA - rolled leaves + stomata on inside = ↓ water loss by transpiration - sunken stomata = build up of layer of humidity = ↓ water loss by transpiration - thick waxy cuticle = ↓ water loss - hairs = trap layer of moist air = ↓ transpiration - wide spreading shallow + deep roots = maximise water absorption - vertical stem = avoid sun on large SA
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Explain how xerophytes are adapted to absorb sufficient water in the conditions in which they live [4]	Root cells have lower water potential Widespread long roots below surface Deep roots to absorb water from the watertable Absorb water by osmosis Many root hairs = large SA						
Explain how xerophytes are adapted to reduce water loss to the atmosphere [3]	Sunken stomata Few stomata Rolled leaves Thick cuticle Hairs on leaves = Reduce rate of transpiration						
Suggest how xerophytes protect themselves against herbivores [2]	Make toxins Spines/needles						
Describe natural selection	There is a mutation. This creates variation in the population. There is production of many offspring. The offspring compete for resources e.g. (name a resource specific to the organism in the exam question!) There is a struggle for survival, those that are better adapted survive and reproduce, passing on the advantageous alleles to their offspring. Over time, the whole population becomes.....						
What is evolution?	the change in adaptive features of a population over time as the result of natural selection						
Define the process of adaptation	the process, resulting from natural selection, by which populations become more suited to their environment over many generations						

Give an example of natural selection	Bacteria becoming antibiotic resistant
Describe how bacteria become antibiotic resistant	There is a mutation. This creates variation in the population; some are killed by antibiotics, and some bacteria are not. There is a struggle for survival. When antibiotics are taken, those antibiotic resistant bacteria that are better adapted survive and reproduce, passing on the advantageous alleles that causes antibiotic resistance to their offspring. Over time, the whole population becomes antibiotic resistant
Describe selective breeding	selection by humans of individuals with desirable features, crossing these individuals to produce the next generation, selection of offspring showing the desirable features
State the similarities between natural and artificial selection	Both require variation in the population, caused by random mutations, both cause changes in the population over time, both affect fitness of the population
Give examples of selective breeding over many generations to serve humans	improve crop plants, domesticated animals e.g. dogs

B19. ORGANISMS AND THEIR ENVIRONMENT QUIZ CARDS

Question	Answer
What is the principal source of energy input to biological systems?	The sun
Define a food chain	showing the transfer of energy from one organism to the next, beginning with a producer
Define a food web	a network of interconnected food chains
Define producer	an organism that makes its own organic nutrients, usually using energy from sunlight, through photosynthesis
Define consumer	an organism that gets its energy by feeding on other organisms
Define herbivore	an animal that gets its energy by eating plants
Define carnivore	an animal that gets its energy by eating other animals
Define decomposer	an organism that gets its energy from dead or waste organic materia
Nematodes feed on dead and decaying material. Explain why this gives nematodes an important role in ecosystems [3]	Recycles nutrients Releases carbon dioxide for photosynthesis Removes organic waste
Describe the role of the bacteria in the root nodules [2]	Nitrogen fixation Converts nitrogen to ammonium ions Converts ammonia to amino acids
What type of energy does the sun provide?	light and heat
What type of energy do living organisms contain?	chemical
Describe how energy gets from the sun to a goat. A goat eats grass.	Radiation comes from the sun. Green plants/algae convert sunlight energy into chemical energy through photosynthesis, which creates biomass. The goat (primary consumer) eats the grass (producer)
Describe how energy is transferred from a mammal to the environment	Some energy is lost as heat, some is returned to the environment via waste products (urine, faeces), some is returned to the environment when the organism dies
In a food chain, how is energy transferred from one organism to another?	INGESTION
A grey kestrel preys on lizards. Lizards consumer praying mantis. A moth consumes grasses. The praying mantis is a predator of moths. Draw a food chain	Grasses -> moth -> Preying mantis -> lizard -> Grey kestrel
Define trophic level	the position of an organism in a food chain, food web, pyramid of numbers or pyramid of biomass
Why is the transfer of energy from one trophic level to another is inefficient	At each trophic level energy is 'lost' (around 90% energy lost): some is used by the organism for growth / movement, maintain a constant body temperature lost to the environment in waste products (faeces and urine) when eaten not all of the organism is consumed (e.g. teeth, bones)
Why do food chains usually have fewer than five trophic levels	Most food chains have fewer than 5 trophic levels. This is because of the energy lost at each level. (Only 10% energy transfer, Energy lost in respiration, excretion, not all parts of organism eaten) The top consumers (apex predators) need to consume a large number of organisms to obtain the energy they require. There are just not enough organisms at each level to sustain an infinite amount of trophic levels.

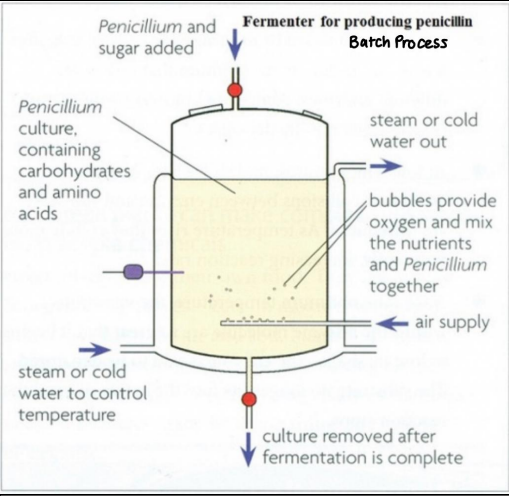
Which is it more efficient to supply plants as human food?	By eating plants there are two trophic levels - producer and primary consumer, so little energy is lost (compared to if animals ingest the grass, and humans eat the animals = more energy is lost through movement / growth of the animal, the animal maintaining a constant body temperature, waste products, not all parts of the animal being eaten) Fewer trophic levels if consume plants Energy lost between trophic levels Only 10% energy is transferred Energy lost by respiration, excretion, not all parts of the organism eaten etc..
Why is it inefficient to feed crops to livestock for human food?	By feeding livestock the plants, energy is lost from the food chain before it reaches the human= there is an additional trophic level
How do humans affect food chains?	over-harvesting, introducing foreign species to a habitat, removing habitats, causing extinctions
Draw a pyramid of biomass	Pyramids of biomass: $\text{biomass} = \text{dry mass} \times \text{no. of organisms}$
Draw a pyramid of numbers	Pyramid of numbers:
Discuss the advantages of using a pyramid of biomass rather than a pyramid of numbers to represent a food chain	Pyramids of biomass are more accurate than pyramids of number as they give a closer approximation of the amount of energy available at each level. It is a representation of the dry mass of organisms at each level rather than the number which can be misleading. Pyramids of biomass are (almost) always pyramid shaped whereas pyramids of numbers are not always.
Explain the advantages of presenting information about food webs as a pyramid of biomass and not as a pyramid of numbers [3]	Pyramid of biomass always pyramid shaped whereas pyramid of numbers not always Biomass indicates how much food is available and is an indicator of the energy available One large individual is shown in the same way as one small individual in the pyramid of numbers
Draw the carbon cycle	

Give 2 reasons why global carbon dioxide levels are increasing	Burning fossil fuels releases carbon dioxide, deforestation releases carbon dioxide (as trees are a store of carbon / carbon sink as take in CO ₂ during photosynthesis)
Draw the water cycle	
Describe the nitrogen cycle	– decomposition of plant and animal protein to ammonium ions by decomposers – nitrification by nitrifying bacteria change ammonium ions into nitrate ions – nitrogen fixation by lightning and nitrogen-fixing bacteria in root nodules – absorption of nitrate ions by plants – production of amino acids and proteins in plants and animals – feeding and digestion of proteins by animals – deamination (removal of amino group from ammonia as it is toxic to form urea) – denitrification by bacteria (change nitrate ions to nitrogen gas)
Draw the nitrogen cycle	
State the roles of microorganisms in the nitrogen cycle	– decomposition of plant and animal protein to ammonium ions by decomposers – nitrification by nitrifying bacteria change ammonium ions into nitrate ions – nitrogen fixation by nitrogen-fixing bacteria in root nodules – denitrification by bacteria (change nitrate ions to nitrogen gas)
Describe how nitrogen in proteins in dead leaves is recycled to be absorbed by plants [4]	Decomposers break down proteins to amino acids Amino acids are converted to ammonium ions Ammonium ions converted to nitrate ions through nitrification Plants absorb nitrate ions
Define population	a group of organisms of one species, living in the same area, at the same time
State the factors affecting the rate of population growth for a population of an organism	food supply, predation, disease

Discuss the increase in human population size over the past 250 years and its social and environmental implications	increase in farming techniques and medical care = increase in global population. This has had many environmental impacts. These include: clearing of natural habitats, and therefore reduced biodiversity, for agriculture and housing, increased consumption of fossil fuels which leads to global warming, increased use of water, over fishing and introduction of invasive foreign species.
Describe the negative effects of deforestation on soil ecosystems [4]	More flooding and waterlogging Soil erosion Loss of habitats = disrupt food chains Extinction of species
Define community	all of the populations of different species in an ecosystem
Define ecosystem	a unit containing the community of organisms and their environment, interacting together
Identify the lag, exponential (log), stationary and death phases in the sigmoid population growth curve for a population growing in an environment with limited resources	Sigmoid curve of population growth: (environment with limited resources) Lag phase Exponential phase Stationary phase Death phase few organisms reproducing population ↑ rapidly NO limiting factors death = birth rate death rate > birth rate due to lack of food etc.. limiting factor present population size for bacteria time

B20. BIOTECHNOLOGY AND GENETIC ENGINEERING QUIZ CARDS

Question	Answer
List examples of biotechnology	Yeast in breadmaking Yeast in ethanol production for biofuel Enzymes in biological washing powders Pectinase to make fruit juice Lactase to make lactose-free milk Using penicillium mould to make penicillin antibiotic Genetic modification of crops Genetic modification of bacteria to produce insulin
Describe why microorganisms e.g. bacteria are useful in biotechnology / genetic modification	1. There is a lack of ethical concerns over their manipulation and growth 2. They share the same genetic code as all other organisms . This means human DNA can be inserted into the bacterial genome and the bacteria will recognise the DNA as its own. It can then produce the proteins which the section of DNA codes for. 3. The presence of plasmids allows for easy insertion of DNA into the bacterial genome. 4. Fast reproduction rate 5. Can make complex molecules
Write the word equation for anaerobic respiration in yeast	Glucose --> ethanol + carbon dioxide
Describe how yeast are used in bread making	Fermentation occurs, which is the anaerobic respiration in yeast (no oxygen). Glucose is broken down in alcohol, carbon dioxide and energy. Carbon dioxide bubbles cause bread to rise
What makes bread rise?	carbon dioxide
Describe how ethanol is made (biofuel)	Crops e.g. maize, or sugar cane are grown. Microorganisms/yeast is used to anaerobically ferment the sugar from the crop, producing ethanol and carbon dioxide
Advantages of biofuels	no production of toxic gas carbon neutral (plants use CO ₂ for photosynthesis and releases CO ₂ during fermentation = balanced CO ₂ conc) renewable IF used at the same rate as production high energy content
Disadvantages of biofuels	destruction of habitat extinction of species of animals/plants biofuel crops grown instead of food crops
Name 4 enzymes in biological washing powders and what each breaks down.	Proteases (protein e.g. in blood, eggs), Lipases (fat e.g. in butter), Amylases (starch), Cellulases (cellulose fibres on the outside of cotton fabrics) Products are soluble so they are washed away in the water
Describe how you could compare the effectiveness of biological versus non-biological washing powders	IV - Biological or non-biological washing powder DV - Use a scale of whiteness to score how clean the fabric is CV - type of stain, volume of solution used to create stain, type of fabric/material, temperature at which washed, length of time of washing R&A - Repeat and calculate an average / remove anomalies and calculate an average S - Wear gloves to avoid contact with biological washing powder (contains enzymes) encase of allergic reaction

Describe how you could investigate the effectiveness of different pectinase concentrations on apple juice	IV - Test different concentrations of pectinase (e.g. 10 different concentrations made by adding 10cm³ to 10cm³ of water, 9cm³ to 10cm³ of water, 8cm³ to 10cm³ of water.... 0cm³ to 10cm of water as a control) added to crushed apple DV - The volume of apple juice produced CV - Type or pectinase, type/species of apple, mass of apple, volume of water, temperature of water bath the apple and pectinase solution are left in, time apple and pectinase solution left for S - Goggles/gloves to avoid pectinase coming into contact with eyes/skin as may cause an allergic reaction, water bath at 40C as optimum for pectinase and so does not burn R&A - Repeat investigation 5 times and calculate an average / remove anomalies and calculate an average
Why are some people allergic to lactose?	Do not produce lactase enzyme so lactose cannot be broken down / digested
Describe how lactose-free milk can be produced	1. lactase made from yeast 2. lactase bound to the surface of alginate beads 3. milk is passed repeatedly over beads 4. lactose is broken down into galactose and glucose 5. immobilised enzyme to be used again and does not affect the quality of lactose-free milk
What is lactose?	A sugar (carbohydrate) found in milk
Describe how you could test the effectiveness of lactose-free milk production	1. measure identical volumes of NORMAL milk 2. add lactase to one 3. no lactase to the other (control) 4. leave overnight 5. use Benedict's solution and heat to test for glucose in both (turns from blue to yellow/orange/brick red) 6. should only be present in the milk enzyme added
Describe how fermenters are used in the production of penicillin	1. fermenter inoculated with penicillium fungus 2. add nutrients, ammonia/urea, vitamin B. 3. Keep at 30C and pH 6.5 4. After 6 days extract penicillin 5. sterilise fermenter and repeat for next batch
what conditions must the fermenter be in to produce penicillin?	constant temperature (30 degrees C) constant pH (6.5pH)
Why does it take 30 hours for penicillin production to begin?	production starts in the post-exponential growth phase of Penicillium: once reactants start to be used up and a limiting factor is present, the fungus' response is to make penicillin. penicillin is the secondary metabolite (=batch culture)
Define fermentation in an industrial context	a process that uses microorganisms to make useful product
Label a diagram of a fermenter	
What is added to a fermenter and why?	nutrients (sugar/starch for respiration), ammonia/urea (N ₂ for protein), vitamin B (respiration)
How is a fermenter sterilised?	radiation, high pressure, high temperatures
What is the function of the stirrer in a fermenter?	to mix everything evenly

What is the function of the air supply in a fermenter?	microorganisms need oxygen to respire
What is the function of a water-cooled jacket in a fermenter?	steam/cold water pumped in to keep temperature cool/constant removes excess heat from respiration
What is the function of a temperature probe in a fermenter?	To monitor temperature to ensure it is optimal for the microorganisms' enzymes / growth
Explain the importance of the water jacket in a fermenter [2]	Maintain constant temperature that is optimum for enzymes Prevent enzymes from being denatured as active site will change shape
Explain the importance of a source of nitrogen in a fermenter [2]	To make amino acids to make proteins (e.g. enzymes)
Explain the importance of the air lock in a fermenter for the anaerobic respiration of yeast [2]	Control pressure Allow carbon dioxide to escape Prevent entry of bacteria
Explain the function of the stirrer in a fermenter [1]	keep microorganisms suspended maintain even temperature ensure microorganisms' access to nutrients
State why the air and nutrients that are added to the fermenter are sterile [1]	To prevent contamination
Explain why ammonia and air are pumped into the fermenter [3]	Ammonia to provide nitrogen for amino acids to make proteins Air to release oxygen for fungus to carry out aerobic respiration that releases energy for growth
Suggest why steam is used to clean fermenters [2]	Prevent contamination Kills pathogens
Suggest why it is important to stir the contents of the fermenter continuously [3]	Uniform mixture Ensure bacteria have more access to the nutrients So oxygen/nutrients dissolve Bacteria absorb substances more easily
Define genetic engineering	Genetic engineering is changing the genetic material of an organism by removing, changing or inserting individual genes
State examples of genetic engineering	– the insertion of human genes into bacteria to produce human insulin – the insertion of genes into crop plants to confer resistance to herbicides – the insertion of genes into crop plants to confer resistance to insect pests – the insertion of genes into crop plants to provide additional vitamins
What are plasmids?	small, circular sections of DNA genes carried in plasmids provide bacteria with genetic advantages e.g. antibiotic resistance
Describe how bacteria can be genetically engineered to produce a human protein e.g. insulin	– isolation of the DNA making up a human gene using restriction enzymes, forming sticky ends – cutting of bacterial plasmid DNA with the same restriction enzymes, forming complementary sticky ends – insertion of human DNA into bacterial plasmid DNA using DNA ligase to form a recombinant plasmid – insertion of plasmid into bacteria – replication of bacteria containing recombinant plasmids which make human protein as they express the gene
Is insulin a primary or secondary metabolite?	primary
List common features of a bacterial cell	plasmid, capsule, flagellum, chromosomes, cytoplasm, cell wall cell membrane, ribosome
Discuss the advantages of genetically modifying crops, such as soya, maize and rice	solving global hunger (crops that can tolerate extreme environments / soils, increasing food production), environmentally friendly (resistant to insects so less pesticide needed), consumer benefits (improved flavour/quality, fewer additives)
Discuss the disadvantages of genetically modifying crops, such as soya, maize and rice	environmental safety (could become invasive weeds), food safety (new combinations of genes may have unknown effects), biodiversity (increasing use of specific GM crops reduces variety of plants)

B21. HUMAN INFLUENCES ON ECOSYSTEMS QUIZ CARDS

Question	Answer
How has modern technology resulted in increased food production?	– agricultural machinery to use larger areas of land and improve efficiency – chemical fertilisers to improve yields – insecticides to improve quality and yield – herbicides to reduce competition with weeds – selective breeding to improve production by crop plants and livestock, e.g. cattle, fish and poultry
Describe the negative impacts to an ecosystem of large-scale monocultures of crop plants	– land clearance / deforestation to allow large-scale monoculture to be planted, reducing habitats for wild species, causing extinction - disrupting food chains - disrupting nutrient cycles - Loss of soil / loss of nutrients / reduced soil fertility / soil erosion - monoculture has little variation, so crop is susceptible to disease - large volumes of water and pesticides used to maintain high crop yield - little variation means no natural selection can take place - monoculture means less biodiversity of species, so less habitats - monoculture means less biodiversity, so more of the same habitat for pest species meaning more pesticides are used - large corporations benefit, not local, small-scale farmers
Describe the negative impacts to an ecosystem of intensive livestock production	large production of methane, unethical conditions (dark, little space, de-beaking chickens), contributing to antibiotic resistance if antibiotics used in animal feed, slurry (urine and faeces from intensive/factory farms) contaminate local water ways/rivers, fish farms with wasted food contaminate the oceans/lakes/disrupt food chains
Discuss the social, environmental and economic implications of providing sufficient food for an increasing human global population	social - large areas of land needed, this can displace local people (sometimes by force), increasingly mechanised farming means more rural employment as less workers needed environmental - large inputs of energy from fossil fuels needed to maintain conditions for animals, large areas of land cleared, contamination of land with pesticides/slurry etc economic - large scale food production by corporations, not local communities, so does not help reduce poverty
Explain how increasing human population leads to food shortages	increased demand for food, unequal global food distribution, war over resources / poverty, limited land for farming due to urbanisation, poor farming practices, high levels of pollution causing crop failure, decreasing soil fertility causing crop failure
What contributes to famine?	unequal distribution of food, drought and flooding, increasing population and poverty
Describe the reasons for habitat destruction	– increased area for food crop growth, livestock production and housing – extraction of natural resources – marine pollution
How can humans have a negative impact on habitats?	deforestation / land clearance for farming / urban development, habitat destruction, extinctions, altering food chains & webs by introducing new species, killing large predators, killing large herbivores, overfishing, overgrazing land with farm animals
Explain the undesirable effects of deforestation on the environment	Extinction of species occurs due to loss of habitat, loss of food, population sizes decrease, possible extinction of species, loss of biodiversity Disruption of food chains / webs Loss of soil is due to erosion. Plant roots help to support and stabilise soil stopping it from being lost in wet weather. Flooding due to reduced water uptake by trees. Large trees absorb anywhere up to 100 gallons of water in a day and return it to the atmosphere. Increase of carbon dioxide in the atmosphere due to less carbon dioxide being fixed in plants during photosynthesis.

What are the reasons for habitat destruction?	1. An increase in the size of the area needed for food crop growth, livestock production and housing 2. The extraction of natural resources 3. Marine pollution																	
List the undesirable effects of deforestation	habitat destruction, to include extinction, loss of soil, flooding and increase of carbon dioxide in the atmosphere																	
State the sources and effects of pollution of land and water	industrial chemical waste – includes variety of toxic substances including lead, mercury and cyanide fertilisers (mostly N and P) herbicides for killing weeds pesticides for killing pests and diseases solid waste (biodegradable and non-biodegradable rubbish) non-biodegradable plastics nuclear fall out (beta and gamma radiation)	factories arable agriculture arable agriculture arable and livestock agriculture domestic and industrial waste packaging atomic bomb, accidents at nuclear power stations, nuclear tests	can be fatal to wildlife and humans; can accumulate in organisms eutrophication in fresh water (rivers, lakes) spray drift kills harmless plants, persist in the environment accumulates in food chains, kills harmless (non-pest) species buried in ground (landfill) or left on rubbish tips. Health hazard, leakage of toxic liquids and release of methane buried in ground, does not decay so takes up space death with high exposure, cancers in humans, mutations in non-human species															
State the sources and effects of pollution of water (rivers, lakes and the sea) by chemical waste	Chemicals from factories can be fatal to wildlife and accumulate in organisms agricultural chemicals from farming e.g. herbicides and pesticides can contaminate the environment, destroy habitats, be ingested by other species, and spread through the food chain (bioaccumulation)																	
State the sources and effects of pollution of water (rivers, lakes and the sea) by discarded rubbish	Discarded rubbish from packaging/human settlements/landfills/rubbish dumps - causes health hazard, leakage of toxic chemicals into the natural environment, release of greenhouse gases if burned/incinerated																	
State the sources and effects of pollution of water (rivers, lakes and the sea) by untreated sewage and fertilisers	From arable land/agriculture/human communities with no sewage treatment infrastructure - causes eutrophication																	
Explain the process of eutrophication of water	– increased availability of nitrate and other ions – increased growth of producers / aquatic plants – increased decomposition after death of producers – increased aerobic respiration by decomposers – reduction in dissolved oxygen – death of organisms/fish requiring dissolved oxygen in water																	
Discuss the effects of non-biodegradable plastics in the environment, in both aquatic and terrestrial ecosystems	They take up space in landfill sites, They are identified as food by some organisms e.g turtles often think they are jellyfish, Animals can become entangled, They release harmful chemicals, toxic liquids, methane and carcinogens.																	
Describe the negative impacts of female contraceptive hormones in water courses	reduced sperm count in men and feminisation of aquatic organisms																	
Name 4 pollutants of air, their sources and effect on the environment	<table><tr><th>pollutant</th><th>sources</th><th>undesirable effects of pollutant on the environment</th></tr><tr><td>sulfur dioxide</td><td>burning fossil fuels</td><td>cause of acid rain</td></tr><tr><td>carbon dioxide</td><td>burning fossil fuels</td><td>enhances greenhouse effect</td></tr><tr><td>methane</td><td>Cattle and paddy fields for growing rice, coal and oil extraction</td><td>enhances greenhouse effect</td></tr><tr><td>nitrogen oxides</td><td>motor vehicles, fertilisers</td><td>cause of acid rain</td></tr></table>	pollutant	sources	undesirable effects of pollutant on the environment	sulfur dioxide	burning fossil fuels	cause of acid rain	carbon dioxide	burning fossil fuels	enhances greenhouse effect	methane	Cattle and paddy fields for growing rice, coal and oil extraction	enhances greenhouse effect	nitrogen oxides	motor vehicles, fertilisers	cause of acid rain		
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Discuss the causes of acid rain	Power stations / vehicles release sulfur dioxides and nitrogen dioxides into the atmosphere, to form sulfuric acid and nitric acid when oxidised in clouds (dissolve in water vapour)																	
Discuss the effects of acid rain	water droplets contain nitric and sulfuric acid, when raining this kills trees/vegetation, decreases pH of lakes/rivers, affects/kills aquatic organisms, erodes limestone and chalk buildings/statues/land formations, causes potassium and calcium in soils to become soluble and they leach out of the soil, reducing soil fertility																	

State the measures that are taken to reduce sulfur dioxide pollution and reduce the impact of acid rain	low sulfur fuels (remove sulfur from coal before using it to reduce sulfur oxides released when burning), flue gas desulfurisation (removes sulfur from power station chimneys by treating waste gases with wet powdered limestone to neutralise the acid), catalytic converters (reduce nitrogen oxides in exhaust fumes of cars)
Name 2 greenhouse gases	carbon dioxide and methane
Why are greenhouse gas emissions increasing?	methane - more cattle farming, rice paddies, landfills with rotting food waste, extraction of oil and gas carbon dioxide - more factories, more burning of fossil fuels to provide electricity, more cars, aeroplanes, more deforestation (trees are a carbon sink - they store it!)
Explain how increases in greenhouse gases cause an enhanced greenhouse effect that leads to climate change	solar radiation passes through the atmosphere and warms the earth's surface short wave radiation from sun passes through the carbon dioxide layer the earth radiates/reflects long wave radiation heat energy (infra-red radiation) back into space. carbon dioxide, water vapour and methane are greenhouse gases that absorb some of this heat energy/radiate it back to the Earth, increasing the earth's temperature. It prevents infrared radiation from escaping from the atmosphere This is the greenhouse effect. With the warming of the earth, ice caps melt, sea levels rise, changes in the frequency of natural disasters occur. This is climate change (long-term changes in climate)
Define sustainable resource	one which is produced as rapidly as it is removed from the environment so that it does not run out
Define the term sustainable development	development providing for the needs of an increasing human population without harming the environment
Explain the need to conserve non-renewable resources	Fossil fuels are non-renewable, and being used at a faster rate than they are produced. Fossil fuels have been a major source of energy since the industrial revolution and despite the fact that countries are making efforts to use more sustainable resources, most of the world is still largely dependent on them. It is important to conserve fossil fuels as they are a finite resource and will one day run out due to the time it takes for them to be produced. As fossil fuels are also the main contributor to greenhouse gases it is important that their use is reduced.
Name two examples of resources that we need to maintain on earth	forests and fish stocks
Explain how forests can be sustained	education of landowners / farmers / governments, legal quotas to set limits of how much / give permits to allow some cutting / which land can be cleared / must be re-planted etc, re-stocking - reintroduce tree species that have previously been cut down
Explain how fish stocks can be sustained	education of governments / fishermen, legal quotas to set limits on how much you can catch / what time of year (to avoid breeding season & catching juvenile fish), re-stock - use fish farms to increase population size of some fish species and reintroduce into the wild
Give examples of products that can be reused or recycled	glass, paper, plastic, metal
Why important to conserve wetlands?	Prevent extinction, maintain biodiversity, provide habitats, provide feeding groups for animals, maintain food chains/webs, maintain nutrient cycles e.g. nitrogen, carbon, water cycles, a sink for greenhouse gases/reduce greenhouse effect, provide resources e.g. food, peat, fuel for local communities, for recreation, for tourism, ethical reasons

Describe how sewage is treated to make the water that it contains safe to return to the environment or for human use	Primary treatment: Remove large solids e.g. paper by filtering through screens. Sedimentation - Organic solids (sludge) settle to the bottom of the tank and are separated from liquid waste Secondary treatment: involves microbes (mainly bacteria) to decompose material suspended in water. For example, 1. Activated Sludge: Air containing bacteria and fungi is pumped through sewage (called sludge) in aeration tanks. Bacteria/fungi aerobically respire, breaking down carbohydrates, proteins and fats to carbon dioxide using enzymes. Urea is broken down to ammonia and then converted to nitrate ions 2. Trickle filter beds: Liquid from primary treatment is sprayed over beds of gravel covered in microbes. The microbes break down the organic matter. Tertiary treatment: Settling: liquid from the secondary treatment passes into large tanks. Any organic material and microbes form sludge, which goes back into the secondary phase OR into anaerobic conditions for microbes to produce methane, which is burned to provide energy for the sewage treatment machines Water Treatment: Treat water with chlorine or ozone to kill microorganisms before water returns to the environment or for human use
Describe and explain methods that could prevent overfishing / maintaining fish stocks	education / awareness, reduced demand (to eat from unsustainable fishstocks) / public pressure / campaigning, steps taken by fisherman voluntarily, (legal) quotas / treaties / licenses / laws / restricted catch weight, ensuring sustainable population size / recovery of, endangered, no-catch zones / nursery zones / protected areas, breeding recovery, limited fishing season for stock recovery / optimizes breeding seasons, financial fines to discourage / punish, poor practice; restocking / captive breeding and release to increases gene pool / number of young / reproductively-viable, fish
Explain why organisms become endangered or extinct	climate change, habitat destruction, hunting, pollution and introduced species
Explain the risks to a species if the population size decreases	If the size of a population drops it makes it much harder for the species to survive. This is because the gene pool is reduced (fewer number of ALLELES). This means the variety of genes and therefore characteristics of the organisms in the species is reduced. Reduced variation makes it more difficult for a species to adapt if there is a change in the environment (natural selection)
Describe how endangered species can be conserved	Monitoring and protecting species Monitoring and protecting habitats Education Captive breeding programmes Seed banks
Explain the advantages of conserving ecosystems	To prevent extinction, maintain biodiversity, prevent disruption of food chains/webs, provide habitats for shelter/breeding, recreation, tourism, education, ethical reasons, aesthetic reasons, reduce greenhouse effect/global warming, maintain nutrient cycles (nitrogen, carbon, water)
What does sustainable development require?	– management of conflicting demands – planning and co-operation at local, national and international levels
Explain reasons for conservation programmes	– reducing extinction – protecting vulnerable environments – maintaining ecosystem functions, limited to nutrient cycling and resource provision, e.g. food, drugs, fuel and genes